



Submersible Sewage Pump Series

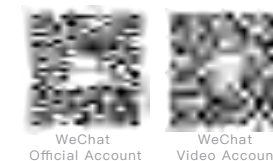


Hangzhou Xizi Pumps Co., Ltd.

Add: No.8,Shunqi Road,Tangxi industrial park,
Linping District Hangzhou City,
Zhejiang province, China

Tel: 0571-88111009/88111688

Web: www.xizipumps.com





Hangzhou Xizi Pumps Co., Ltd. has been dedicated to the research&development and manufacturing of submersible sewage pumps and related systems for over thirty years. Since its establishment in 1994, the company has consistently adhered to the business philosophy of "Customer First, Quality Foremost." We focus on perfecting every product and are committed to providing superior pump products and services for every partner.

The company has collaborated with relevant universities to build a professional R&D and technical team, obtaining over 40 national utility/invention patents and computer software copyrights. In 2022, Xizi Pump was among the first to receive the "National Grade One Energy Efficiency" certification for submersible sewage pumps. Thousands of Xizi pump products are successfully operating worldwide, delivering excellent Xizi solutions in fields such as wastewater treatment, sewage transportation, flood control and drainage, water conservancy projects, irrigation, and water supply and drainage.

Since its inception, Xizi Pump Industry has emphasized the learning and application of modern scientific management methods, having successively obtained ISO9001, ISO14001, and ISO45001 management system certifications. It has also been recognized for multiple years as a "National High-Tech Enterprise," "Zhejiang Provincial Specialized, Refined, Unique, and New Enterprise," "Zhejiang Manufacturing Certification," and "Hangzhou Enterprise Technology Center."

Honor & Qualification



Factory Overview



Focusing on sewage pump and system R&D and manufacturing for 32 years



The factory is located in the Tangqi Industrial Park, Linping District, Hangzhou, with a construction area of 30,000 square meters. It houses comprehensive production, manufacturing, R&D, and testing centers. Key functional areas include metal processing, motor manufacturing, pump assembly, pump station and sewage lift workshops, and a "National Grade One Precision" pump testing center. The factory can customize personalized products and services for clients with diverse needs, providing optimal pump and system solutions.

Professional
Professional dedication, with thirty years of R&D and manufacturing experience in sewage pumps and systems.

High-quality
Possesses a strict quality management system and advanced industry detection equipment, including a water performance laboratory. 100% of products are tested and qualified before leaving the factory.

Efficient
Efficient products, efficient production, and efficient service. Products utilize hydraulic components based on internationally advanced hydraulic models and high-efficiency motors, resulting in wide high-efficiency ranges, high efficiency, and stable performance.

Innovative
Boasts a professional R&D and technical team with numerous national invention patents.

Customization
A complete production center and after-sales team enable the customization of personalized products and services for different customer needs, providing the best pump and system solutions.



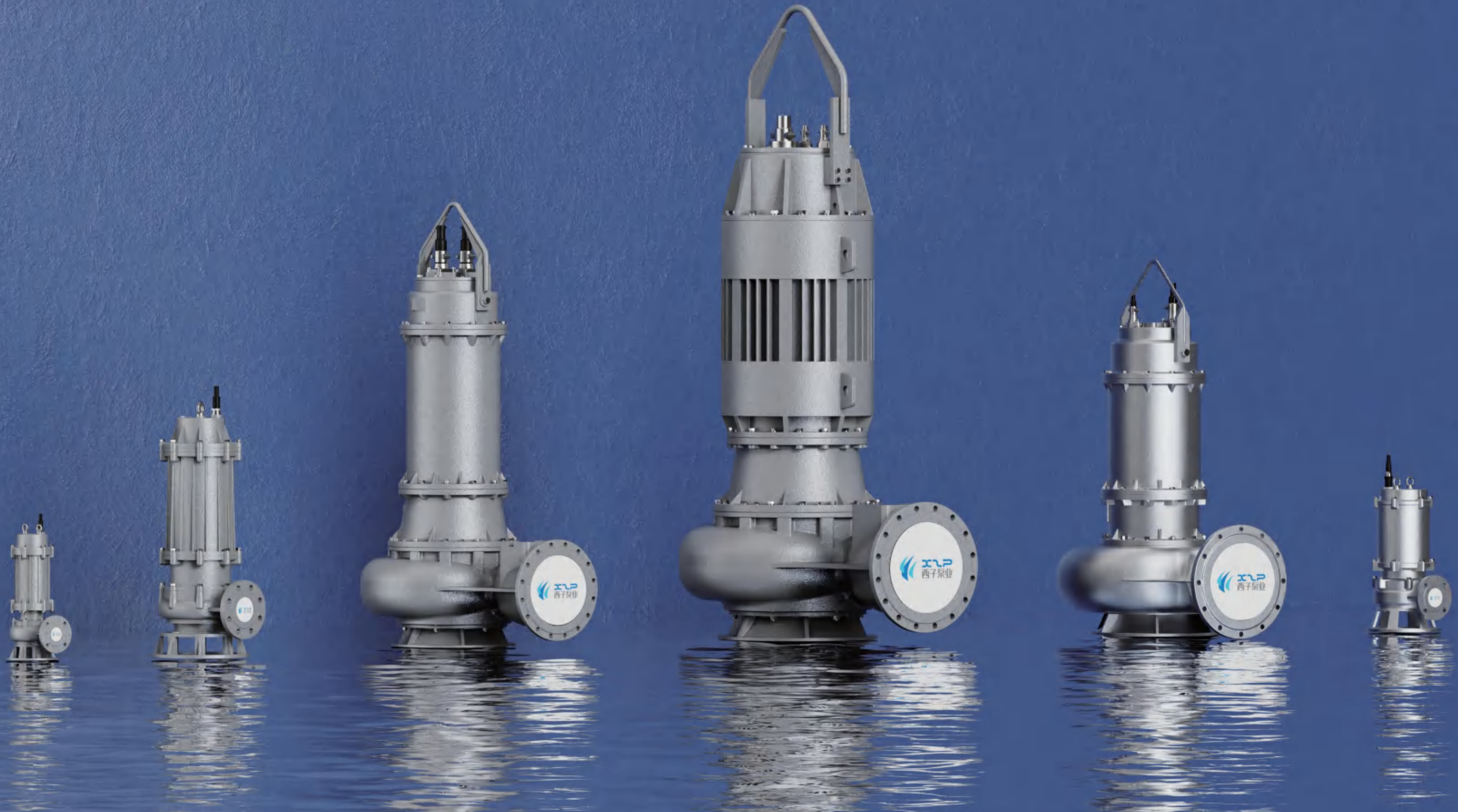


Catalogue

Xizi pumps submersible sewage pump series overview	P. 1–9
WQA high-efficiency submersible sewage pump	P. 10–27
WQE economic submersible sewage pump	P. 28–41
WQF stainless steel submersible sewage pump	P. 42–53
WQA–QG cutting submersible sewage pump	P. 54–64
Ordering, Installition and After-sales service guidelines	P. 66–71



Expert In Energy-Efficient
Submersible Sewage Pumps



XIZI PUMPS

Submersible Sewage Pump Series Overview

For 30 years, Xizi Pumps has been dedicated to the research and development as well as manufacturing of submersible sewage pumps. Our pump products have been successfully applied in various industrial and municipal fields around the world. With the aim of providing better reliability and higher unit efficiency to our customers, we have, on the basis of summarizing over 30 years of experience in submersible pump production and manufacturing, drawn on and introduced advanced international hydraulic model design technology, and carried out comprehensive optimization design in mechanical structure, sealing, protection and other aspects. We meet the highest demands of customers in terms of efficiency, service life, maintenance and economic benefits. The high standards of Xizi submersible sewage pumps are based on decades of hydraulic design experience and extensive professional knowledge. Our goal is to provide customers with world-class pump products and services.

Xizi submersible sewage pumps are divided into four series:

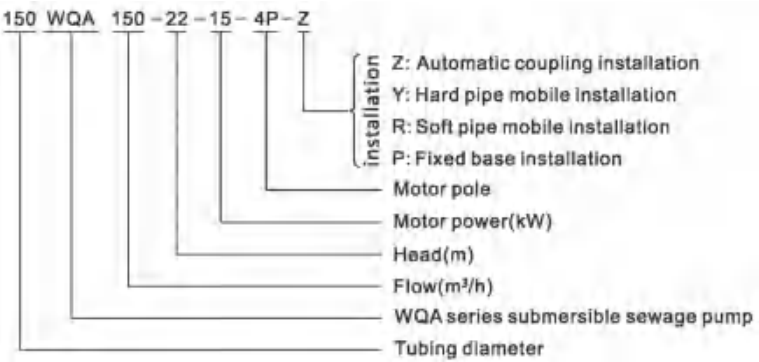
- **WQA** : High-efficiency submersible sewage pump
- **WQE** : Economic submersible sewage pump
- **WQF** : Stainless steel submersible sewage pump
- **WQA-QG** : Cutting submersible sewage pump

The products comply with the national standard GB/T24674 "Submersible sewage and sludge pumps".

Scope of Application:

Municipal engineering, sewage treatment, building construction, water conservancy projects, industrial wastewater discharge, environmental remediation, and other fields.

Model definition:



Product Features:

- The impellers of the WQA/WQE/WQF models all adopt a closed structure, with the main forms being: ① Double-channel type ② Blade type
The double-channel impeller has good performance in passing through debris; the blade-type impeller is energy-saving and has high hydraulic efficiency.
The impeller of the WQA-QG model is of a semi-open structure.

- The pump unit has a compact structure and short shaft extension to reduce vibration and noise. It has undergone strict dynamic and static balance testing to reduce vibration value to the lowest point, and extend the expectancy life of bearing and mechanical seal.

- The cable adopts rubber flexible cable with good oil resistance and excellent mechanical performance, which can work continuously for a long time under 40°C. The interception capacity of the cable has sufficient margin to make the service life longer. The oil-resistant rubber flexible cable meets national standards of JB/T8735.2 and GB/T5013.4.
The water-jet cutting cable prevents accidental water immersion in motor cavity. The pump end cable should be properly fixed to avoid dragging of the sealing device of lead-out wire of the motor upper cover cable.

- The motor adopts high-performance squirrel-cage induction motor, which is specially designed and manufactured for submersible pumps. It implements GB755 standards. Protection class: IP68. Insulation class: F (standard), H (optional)

- The shaft seal adopts single-end mechanical seal to completely isolate motor and pump. It is installed in series up and down to improve sealing reliability, and ensure the long-term and stable operation of pump. The designed service life of mechanical seal is 10,000 hours.

- The bearing adopts lifelong maintenance-free rolling bearings. The upper bearing is deep groove ball bearing or cylindrical roller bearing, which is mainly used to bear radial force. The lower bearing is mainly used to bear radial force and axial force. Given that various pump models will generate different radial force and axial force, some are designed as a double-row angular contact ball bearing, and some are combinations of a diagonal contact ball bearing and a cylindrical roller bearing or deep groove ball bearing to ensure sufficient load margin, making the pump unit run smoothly. The designed service life of the bearing is 50,000 hours.

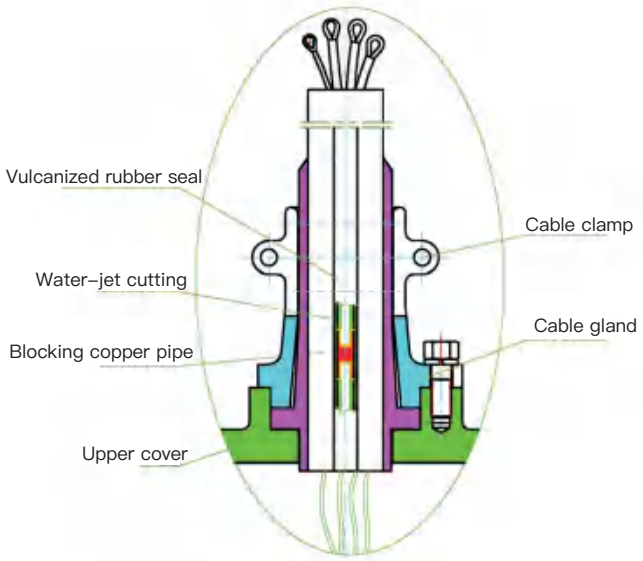
- Oil chamber: Adopt 32# engine oil which can lubricate mechanical seal and cool down the bearing. Executive standard is GB443-89. Add oil till it overflows from the filling hole to ensure that there is a certain gap in the oil chamber, so that the pressure will not rise significantly when oil temperature rises. This is to protect the mechanical seal in oil chamber, thereby protecting the motor.

- Temperature measuring element (thermal protector is equipped for 3kW and below): Each phase winding is pre-embedded with temperature measuring element PTC (115°C). When the pump is in abnormal working condition, and the motor winding temperature rise reaches the setting value, the temperature measuring element will work, the "overheating" indicator light on control cabinet will be turned on, the pump will stop working. This reminds the operator to check and find out overheating causes and solutions in time. The pump can recover by itself after the winding temperature drops.

- Water leakage probe: The water leakage probe is used for leakage detection. When the mechanical seal or the sealing at impeller side is leaking, and the water amount in oil chamber is large to a certain extent, the two electrodes of the probe will be turned on and an alarm signal will be issued (the indicator lights on), to remind the operator to check mechanical seal in time and check whether the oil in oil chamber need to be replaced.

- Float switch: The float switch is installed in the empty cavity next to the bearing cavity on the lower side of motor. It is used to check whether the mechanical seal in oil chamber is damaged and whether the motor seal is leaking. When the water amount in the float switch cavity reaches a certain level, the float switch will float and send an alarm signal through control cabinet (the indicator lights on), to

Structural diagram of water-jet cutting cable



Technical descriptions:

- Xizi submersible sewage pumps can be equipped with flange coupling which implements GB/T17241.6-2008 standard. The standard configuration for coupling outlet flange DN250 and below is PN6 and for DN300 and above is PN10. Pump outlet flange pressure rating is detailed in related pump page.

For WQA/WQF submersible sewage pump 7.5kW and below, pump can be equipped with stirring (JY)
Some models of WQE can be used as stirring pumps. For details, please refer to the sample.

- The standard motor rated voltage of submersible sewage pump is 380V, and the rated frequency is 50Hz.
① 11kW and above can be customized with frequency conversion electric pump unit.
② The 60Hz electric pump unit can be customized.
③ Electric pump units with three-phase voltage below 690V can be manufactured.
④ Single-phase 220V can be manufactured for the pumps 2.2kW and below. (except WQE)
⑤ Frequency conversion can be used for the pumps 7.5kW and below.
⑥ Three-phase 220V can be manufactured for the pumps 3kW and below.

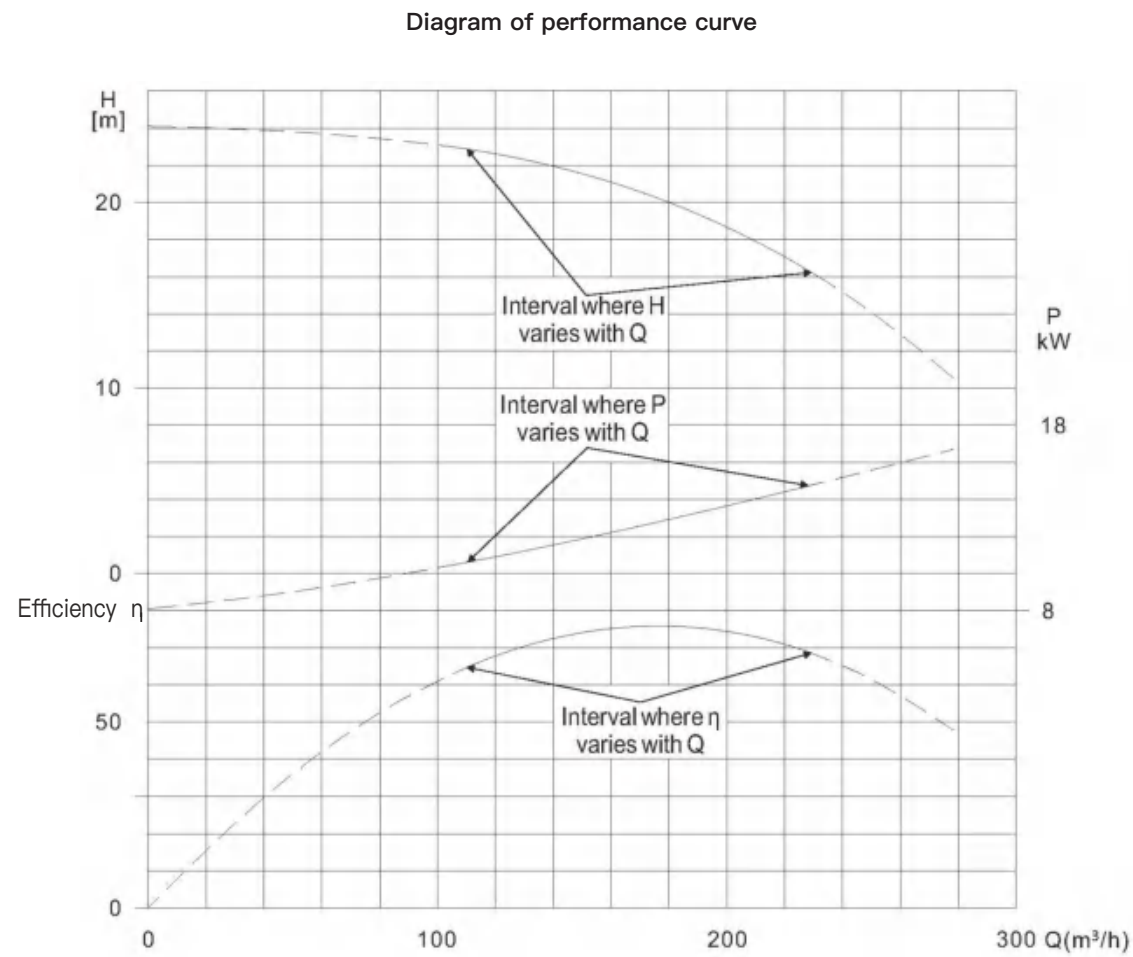
- Wiring method and starting method of motor winding lead wire
① The Y-shaped wiring method is used for 3kW and below. The stator lead-out wire and **a main cable** have been connected in the wiring cavity with this method. It can use direct start.
② The internal delta (Δ) wiring method is used for 4-315kW. When the pump leaves the factory, the winding wires and **1, 2, 3, and 4 main cables** have been connected in the wiring cavity according to power level. Based on the power grid capacity and load permit on customer's site, direct start, external electronic soft starter start or auto-coupling step-down start can be adopted accordingly. (Please refer to wiring diagram for the number of main cable)
③ Single-phase 220V electric pump (0.55-2.2kW) adopts capacitor operation, double capacitor + centrifugal switch start.
④ If there are other requirements (eg: Star Delta Start Y-), please specify separately when placing an order.

- Pump rotation direction
The impeller rotates **counterclockwise** looking from pump inlet. The impeller rotates **clockwise** looking from the motor.

- When the liquid level difference is large on customer's site, use two float switches to control the start/stop liquid level for main pump or large pump, while using one float switch for small pump or standby pump under super high liquid level.

- The terminal box can be installed on the spot when electric control cabinet is far away. When threading pipe (user owned) needs to be laid, confirm the inner diameter of the threading pipe according to the outer diameter of the cable.
7. When the liquid level difference is large on customer's site, use two float switches to control the start/stop liquid level for main pump or large pump, while using one float switch for small pump or standby pump under super high liquid level.
8. The terminal box can be installed on the spot when electric control cabinet is far away. When threading pipe (user owned) needs to be laid, confirm the inner diameter of the threading pipe according to the outer diameter of the cable.

Diagram and description of performance curve



- The solid line part of the curve in the figure represents the reasonable pump operation range where pump efficiency is relatively high and economical.

When pump is running below the leftmost side of the solid line:

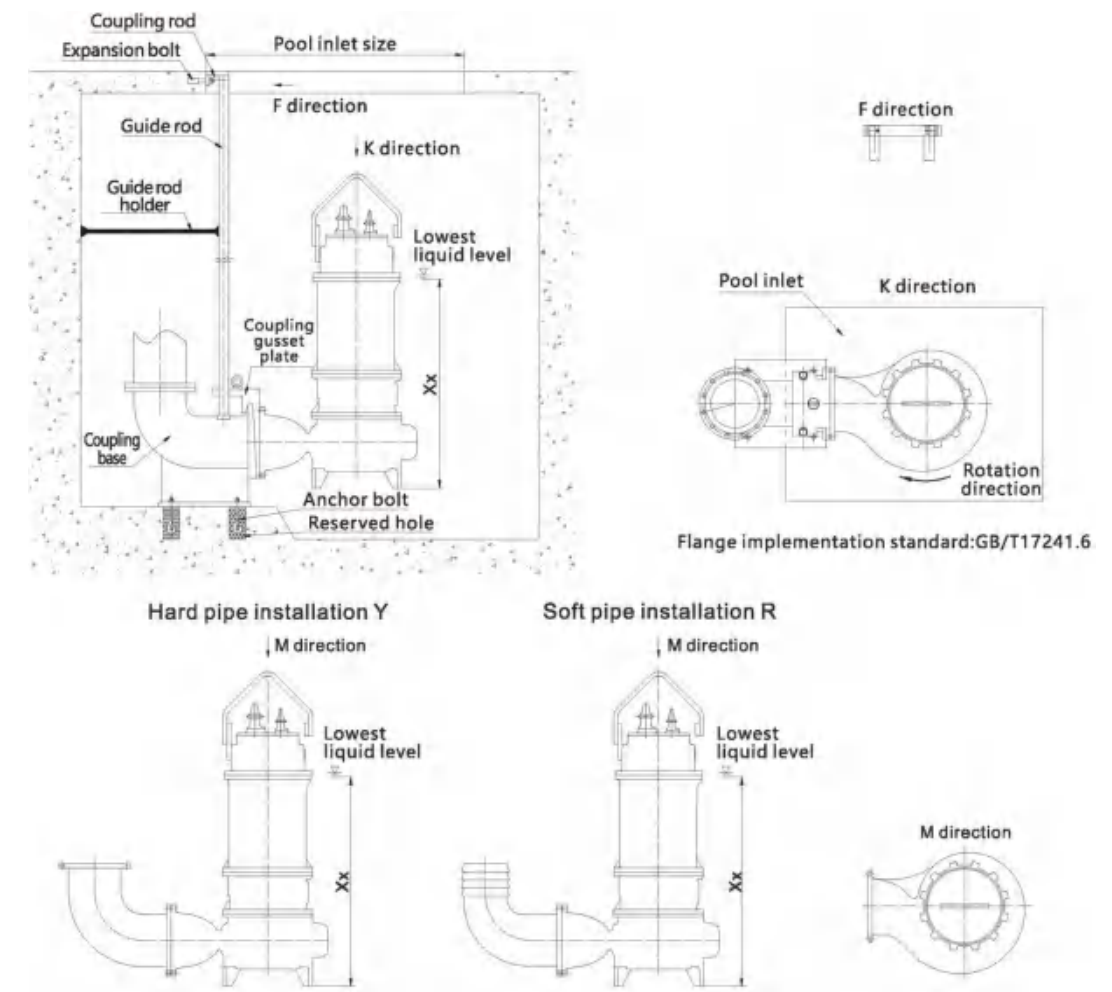
- ① Under long-term low-flow operation, the pump will be always in a state of holding pressure with high pump temperature, excessive vibration and shortened life. When the deviation from the reasonable working range is large, the pump will be damaged to a certain extent after long-term operation.
- ② Large radial force will be generated, bearing and mechanical seal damage is accelerated, rolling keys or even shaft breakage risk will be caused.
- ③ The efficiency will drop and electric energy will be wasted.

When pump is running higher than the rightmost side of the solid line:

The pump will have problems like vibration, noise, cavitation, even over-current and motor burning, etc.

Installation dimension drawing

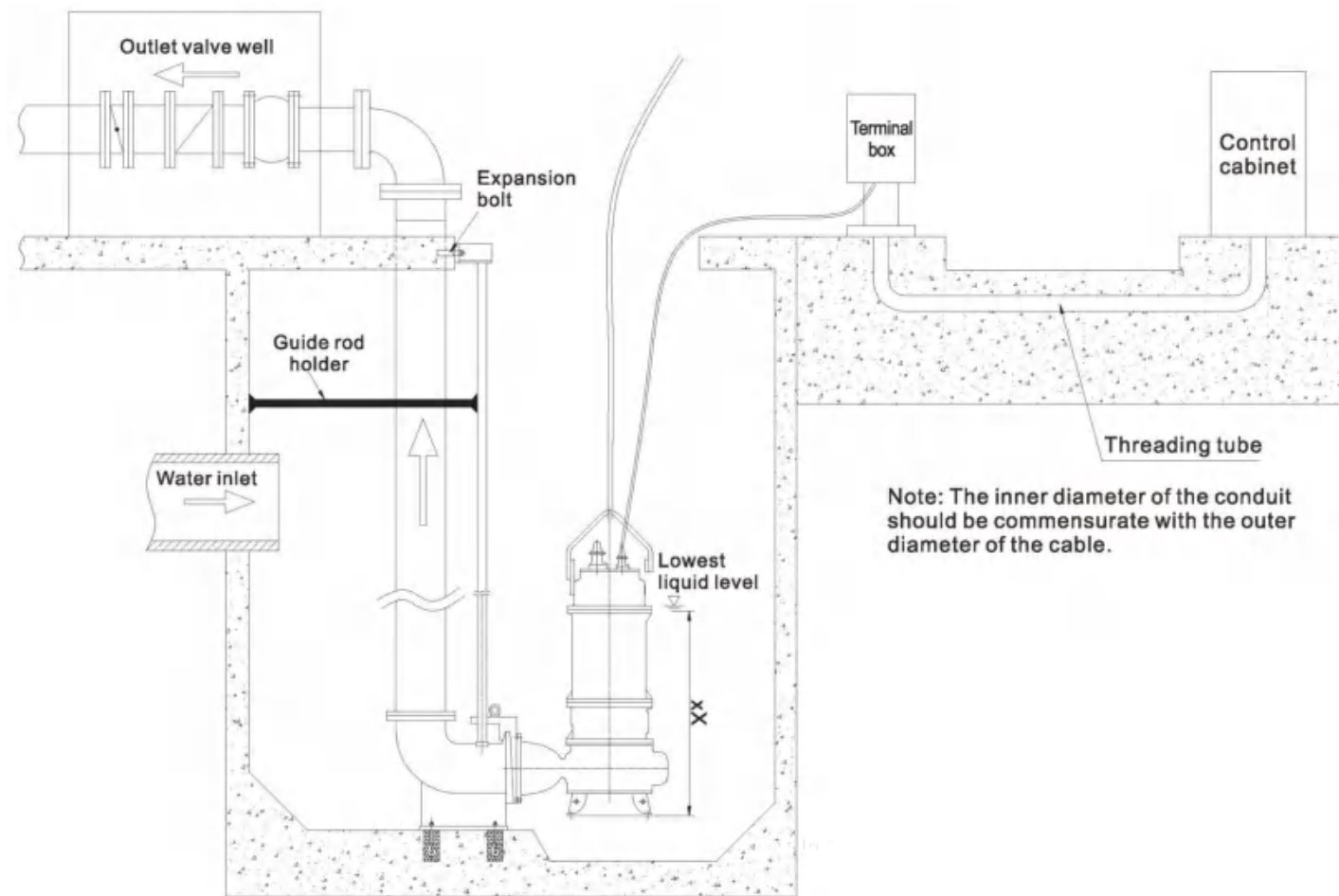
Dimension drawing of coupling installation



- The coupling railings are all fixed with expansion bolts which implement JB/ZQ4763-2006 standard.
- When the length of guide rod is bigger than 5 meters, it is recommended that the user fix the guide rod every 3 meters.
- Calculate guide rod length according to the "pool depth" shown on the figure, and refer to the attached table for the length.
- ∇ indicates the minimum liquid level of pump operation. The pump operation liquid level should be higher than the minimum liquid level. It is best to submerge the pump completely under allowable site conditions, so that the motor can be fully cooled by the medium. The minimum liquid level can be controlled by float switch, ultrasonic level gauge, etc.
- The dimension of coupling base and pump outlet flange shall comply with GB/T17241.6 standard.

Long-distance layout plan and cable specifications

Long-distance layout plan: When the control cabinet is far away from pump room, a terminal box can be installed, which makes the operation more convenient.



Installation method

● Fixed base installation (P):

Fix the base on the foundation, fix the base with anchor bolts, and connect the outlet pipe. It is inconvenient to repair the single pump after running;

● Mobile installation (Y, R):

It is used for first aid or maintenance and construction, flexible and convenient to operate. It can be operated by connecting a hose or a hard pipe with the pump base support. It can also be suspended for use with a sufficiently rigid drainage pipeline.

● Automatic coupling installation (Z):

The coupling device includes: ① Coupling base, ② Coupling gusset, ③ Guide rod, ④ Coupling pull rod.

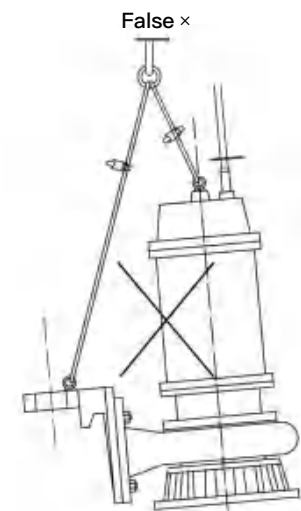
Automatic coupling installation: The coupling device links the pump and pipeline without fasteners, making the connection and disconnection of the pump and coupling base simple.

Operation essentials:

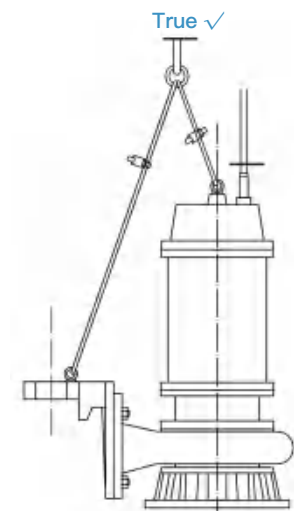
- ① Lay the cement foundation and reserve the anchor bolt holes according to the installation drawing.
- ② Fix the coupling base, coupling rod and guide rod, and the coupling base needs to be poured twice.
- ③ Connect the coupling gusset plate and pump body flange with bolts. Whether to install or lift the pump, you only need to lift the lifting chain and let the gusset plate semi-circular hole slide along the guide rod.
- ④ When installing with a coupling device, the pump must be adjusted to a vertical position using short chains. The lifting point must ensure the pump moves vertically up and down before hoisting can be performed. (As shown in the schematic below.)



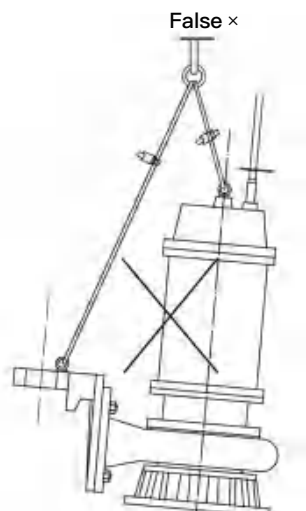
Installation structure diagram of automatic coupling



This situation can't lift pump



Adjust the pump to a **Vertical State** using short chains.

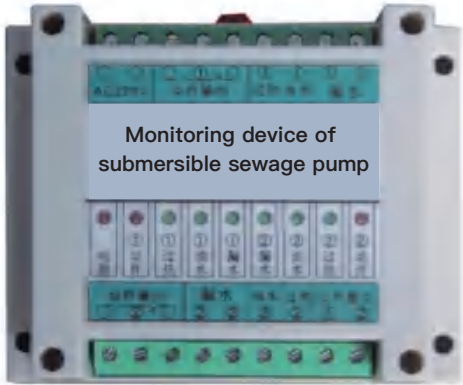


This situation can't lower pump

Special monitoring device of submersible sewage pump

Purpose and function

XIZI submersible pump monitoring device (comprehensive protector) is an optional accessory specially designed for submersible pump protection device by our company, to realize the comprehensive protection function of the submersible pump. It has monitoring and protection functions of motor stator temperature overheating (overheating), oil chamber water leakage (oil and water),wiring cavity and motor cavity water leakage(water leakage). It can monitor 2 pumps at the same time.



Specifications and technical parameters

- ① Specifications
Length × width × height: 115×90×40
Installation size: 105×70,4– φ5; or install on the rail
- ② Working environment
Ambient temperature: −20℃~50℃; Relative humidity≤85%RH; There is no corrosive gas in the surrounding environment
- ③ Power supply: AC220V / 50Hz

Electrical principle and terminal wiring diagram

- 220V access "AC 220V"
- This product can monitor two water pumps at the same time. You can choose to connect to port ① or ② when there is only one pump.
- The common wires of the "overheating", "oil water", "leaking" lead-out wires and the PCC lead-out wires, form the respective protection and monitoring circuits (Refer to schematic diagram).
- "Action output" can be connected to "NO output" or "NC output" according to customer site.

Points for attention.

The inside of the monitoring device is full of electronic circuits, and the insulation resistance value cannot be measured with a

Parameter table of submersible motor

Parameter table of submersible motor (380V / 50Hz)								
No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
1	0.55	2	1.8	0.82	73	8.1	1.87	2.2
2	0.75		2	0.83	75	10.9	2.53	2.2
3	1.1		2.5	0.84	77	17.4	3.71	2.2
4	1.5		3.6	0.84	79	23.4	5.03	2.2
5	2.2		4.9	0.85	81	32.8	7.38	2.2
6	3		6.4	0.87	83	46.1	9.96	2.2
7	4		8.2	0.88	85	58.7	13.24	2.2
8	5.5	2	11	0.88	86	80.3	18.09	2.2
		4	12	0.83	85	80.5	36.4	2.3
9	7.5	2	15	0.88	87	106.5	24.66	2.2
		4	16	0.84	87	107.1	49.63	2.3
10	11	2	22	0.89	88	156.8	35.81	2.2
		4	23	0.84	88	155.4	71.85	2.2
		6	25	0.78	87.5	154.7	108.2	2
		8	26	0.75	86.9	164.3	143.9	2
11	15	2	29	0.89	89	209.3	48.84	2.2
		4	30	0.85	89	223.5	97.97	2.2
		6	31	0.81	91.2	225.6	147.2	2
		8	34	0.76	88	219.8	196.2	2
12	18.5	2	35	0.9	90	254.3	60.24	2.2
		4	36	0.86	90.5	270.8	120.4	2.2
		6	37	0.83	90.7	275.9	180.7	2
		8	42	0.76	88.6	264.7	242	1.9
13	22	2	42	0.9	90	303.8	71.38	2
		4	43	0.86	91	319.5	143.2	2.2
		6	44	0.83	91.2	331.5	214.9	2
		8	48	0.78	89.1	308.9	287.8	1.9
14	30	2	58	0.93	90.3	364.2	97	2.2
		4	60	0.86	92	411.9	194.7	2.2
		6	61	0.84	91.5	408.8	292	2
		8	63	0.79	91	415.8	389.8	2
15	37	2	70	0.88	93.7	512.2	119.6	2
		4	71	0.87	92	501.2	239.6	2.2
		6	73	0.86	91.5	510	360.6	2.1
		8	78	0.79	91.5	502.9	480.7	2.0
16	45	2	82	0.91	92.3	622.2	145	2
		4	85	0.87	92.8	604.8	290.8	2.2
		6	86	0.86	92.5	597.8	438	2.1
		8	94	0.79	92	610.5	584.7	2.0
17	55	2	101	0.9	92.3	758.5	176.9	2
		4	104	0.87	92.6	741.6	355.8	2.2
		6	105	0.86	92.4	723.1	535.4	2.1
		8	114	0.81	92.3	728.7	714.6	1.8
18	75	4	140	0.87	93.1	993.6	483.6	2.2
		6	147	0.86	93.2	981.4	726.3	2
		8	152	0.81	92.8	977.5	974.5	1.8
19	90	4	167	0.89	93.5	1050	580	2
		6	172	0.87	93.5	1008	870	1.7
		8	180	0.82	93.1	1090	1161	1.9

No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
20	110	4	201	0.88	94.5	1200	710	1.8
		6	206	0.86	94.5	1230	1066	1.7
		8	217	0.82	94.1	1324	1420	1.9
21	132	4	241	0.88	94.8	1395	850	1.8
		6	244	0.87	94.5	1440	1280	1.8
		8	260	0.83	93.3	1927	1703	1.5
22	160	4	286	0.9	94.5	1800	1030	2.1
		6	294	0.88	94	1965	1550	1.8
		8	314	0.83	93.3	2204	2064	1.5
23	185	4	333	0.9	94.6	1947	1191	2.0
		6	340	0.88	94	2204	1792	1.6
		8	362	0.83	93.8	2625	2380	1.5
24	200	4	359	0.9	94.6	2112	1288	2
		6	368	0.88	94	2484	1937	1.6
		8	390	0.83	93.9	2835	2574	1.5
25	220	4	396	0.89	95	2758.7	1419.5	2.2
		6	404	0.88	94.2	2812	2132	1.8
		8	430	0.83	93.9	3118	2831	1.6
26	250	4	445	0.9	95	2664	1607	1.7
	250	6	460	0.88	94.4	3105	2424	1.8
	250	8	488	0.83	94	3548	3218	1.6
27	280	4	498	0.9	95	2988	1800	1.7
	280	6	510	0.89	94.4	3428	2715	1.8
	280	8	548	0.83	94	3975	3604	1.6
28	315	4	560	0.9	95	3360	2025	1.7
	315	6	570	0.89	94.5	3735	3054	1.8
	315	8	615	0.83	94	4298	4457.4	1.6

Parameter table of submersible motor (Single phase 220V / 50Hz)								
No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
1	0.55	2	5.2	0.92	70	15	1.88	1.7
2	0.75		5.4	0.92	72	20	2.56	
3	1.1		7.8	0.95	75	30	3.73	
4	1.5		10.2	0.95	76	45	5.1	
5	2.2		14.7	0.95	77	65	7.42	

Parameter table of submersible motor (Three phase 220V / 50Hz)								
No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
1	0.55	2	3.2	0.82	73	14.03	1.87	2.2
2	0.75		3.5	0.83	75	18.88	2.53	
3	1.1		4.4	0.84	77	30.14	3.71	
4	1.5		6.3	0.84	79	40.53	5.03	
5	2.2		8.5	0.85	81	56.81	7.38	
6	3		11	0.87	83	79.85	9.96	

WQA high-efficiency submersible sewage pump ►

WQA

High-efficiency Submersible sewage pump



"National Grade One
Energy Efficiency"
submersible sewage pump

WQA Submersible Sewage Pump is an ultra-high-efficiency submersible sewage pump specifically developed for the wastewater industry. Utilizing internationally advanced hydraulic models, most models achieve "National Grade One Energy Efficiency" making it a true full-head pump that thoroughly solves the problem of overcurrent for customers. At the same time, comprehensive optimization has been carried out in mechanical structure, sealing, and protection, and high-quality motor materials are used to ensure unit stability. The WQA full series includes 182 models.

Customization Available According to Different Working Conditions:
Stainless steel material (WQAF) / Special anti-corrosion coating for seawater
Internal circulation cooling submersible sewage pump / External cooling jacket sewage pump / High-pressure pump

Scope of Application:

Municipal engineering, sewage treatment, building construction, water conservancy projects, industrial wastewater discharge, environmental remediation, and other fields.

Operating Conditions:

Flow Rate: 7 ~ 4200 m³/h
Head: 7 ~ 90 m
Medium Temperature: T ≤ 40°C
Medium Density: ≤ 1100 kg/m³; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 10
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

Product Advantages:

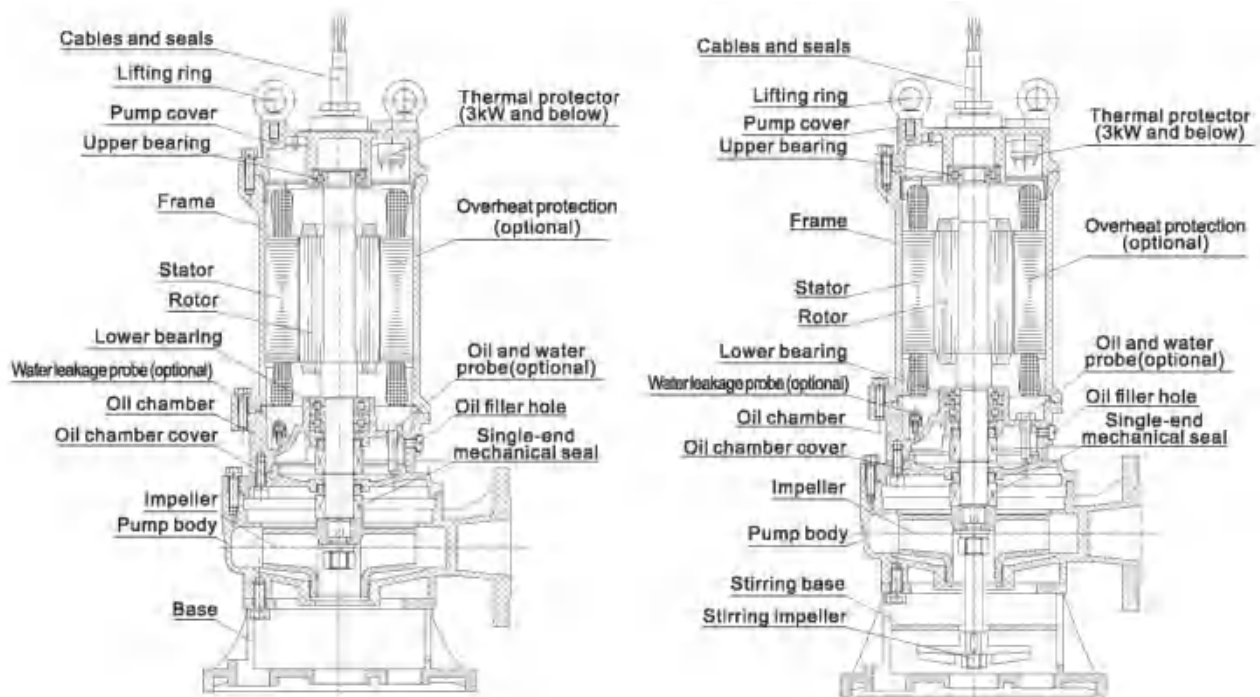
- "National Grade One Energy Efficiency": Pump efficiency exceeds the national standard by 2%–13%. Compared to most domestic products, it can achieve a lower power rating under the same flow and head conditions, resulting in more efficient and energy-saving operation.
- True Full-Head Pump: Wide high-efficiency range and stable performance.
- Water-Cut Process Cable: Effectively prevents liquid water and moisture from entering the pump interior.
- Multiple Detection and Protection Systems: Overheat protection switch, oil-water probe, float switch, all capable of real-time detection, and can achieve functions such as alarm, shutdown, and fault signal retention.
- Motor Inspection Port: Allows inspection of the motor cavity without disassembling the machine.
- Anti-Reverse Rotation Design: Impeller nut is designed with an anti-reverse rotation structure.
- Flexible Lifting Handle: Uses a tripod-style lifting method, with a handle that rotates flexibly for convenient, safe, and secure handling.
- Comprehensive Model Coverage: Easy and convenient selection. Can be flexibly customized according to different needs, suitable for various industries and working conditions.
- 100% Tested and Qualified Before Shipment: Complies with GB/T 12785.



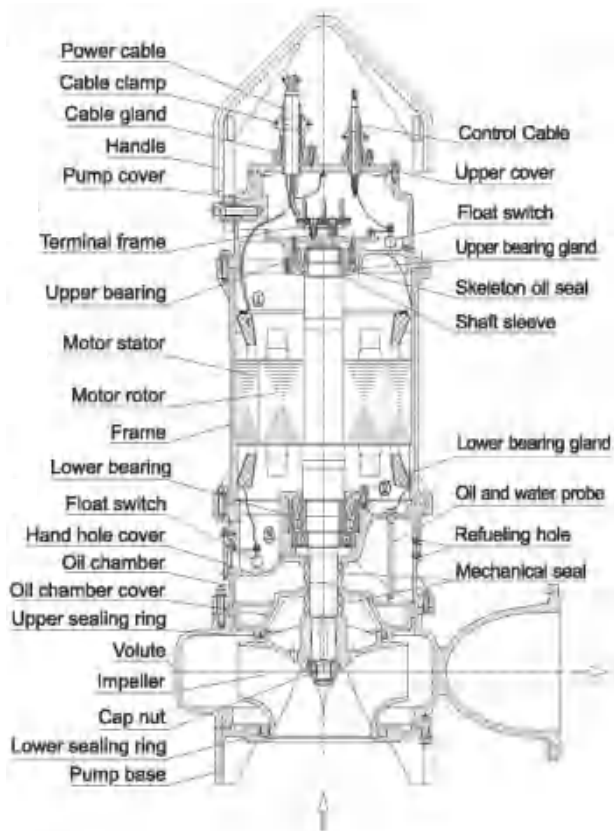
External cooling jacket
submersible sewage pump

Internal circulation cooling
submersible sewage pump

Structure diagram of WQA products



Structure diagram for 7.5kW and below



Structure diagram for 11kW-4P and above

Main materials and configuration table

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		HT200	/
2	Impeller		HT200	304
3	Oil chamber cover		HT200	/
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		HT200	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		2Cr13	304
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/SiC (Q1Q2)	SiC/WC (Q1U1)
11		Spring	Stainless steel	/
12		Rubber parts	(NBR)	(FKM)
13	O-ring		(NBR)	(FKM)
14	External leakage fastener		A2-70	A4-70
15	Protection device (optional)	0.55-7.5kW	Thermal protector for 3kW and below	Water leakage probe/Float switch /Overheating protection

7.5kW and below

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		HT250	QT
2	Impeller		HT250	304/ QT
3	Oil chamber cover		HT250	QT
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		HT200	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		2Cr13	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/WC (Q1U1)	SiC/SiC (Q1Q2)
11		Spring	Stainless steel	/
12		Rubber parts	(NBR)	(FKM)
13	O-ring		(NBR)	(FKM)
14	External leakage fastener		A2-70	A4-70
15	Protection device	11-15kW Optional 18.5-315kW Standard	Water leakage probe/Float switch /Overheating protection	Bearing temperature and vibration sensor

11kW and above

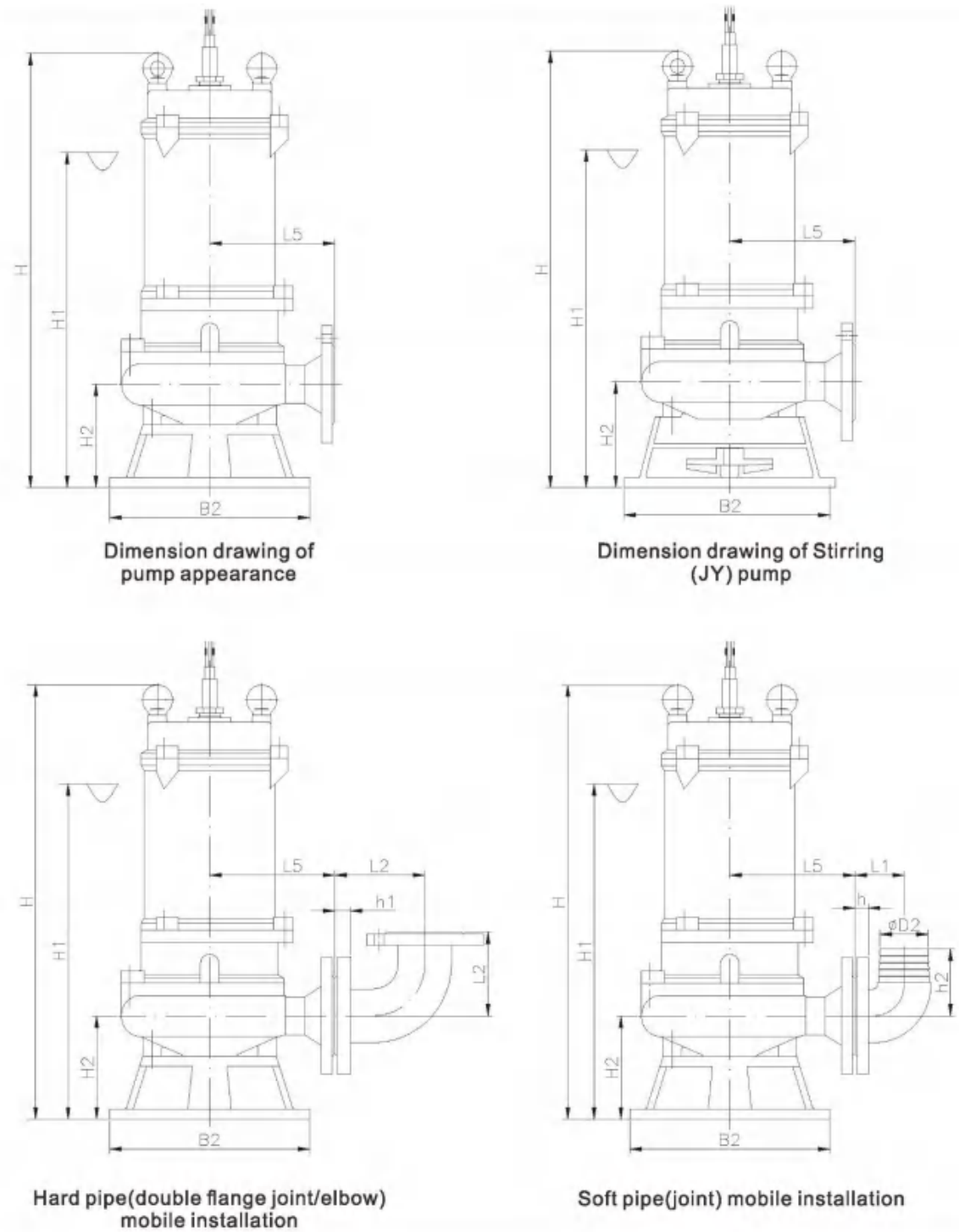
Installation dimension drawing

The technical drawings illustrate the pump assembly with the following dimensions and labels:

- Side View (Left):** Shows the pump's profile with dimensions L10 (Minimum pool inlet), L3, L4, L6, L5, H3, H5, H6, H7, H1, H2, and H4. It also indicates the 'F direction' (Flow direction) and 'K direction' (Shaft direction).
- Top View (Right):** Shows the pump's footprint with dimensions L1, L2, L, and M. It labels the 'F direction', 'K direction', and the shaft diameter $\Phi d \times t$.
- Front View (Bottom Right):** Shows the pump's front with dimensions A, A1, B, B1, L9, L8, L7, J, and N1- $\Phi d1$. It labels the 'Pool inlet', 'K direction', and 'Rotation direction'.
- Other Labels:** 'Lowest liquid level' is indicated on the side view, and 'GB/T17241.6 standard' is noted for the flange.

Model	Flange Connection Size			Coupling Base Size						Adjust according to pump model																		
DN	D	D1	N1-Φd1	A	A1	B	B1	J	n2-Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φd×t				
40	130	100	4-Φ14	108	65	138	68	68	4-Φ15	195	Adjust according to pump model	120	14	150	225	185	70	35	Φ14	65	125	85	13	Φ33×3				
50	140	110		197	168	208	130	120	250	165					265	215	105	42	Φ14	66	135	90	15					
65	160	130		238	190	252	155	135	265	170		280	235		125	50	Φ14	70	155	85	18							
80	190	150	4-Φ19	255	220	225	153	125	310	190		16	/		315	265	145	53	Φ14	80	155	100	18	Φ48×3				
100	210	170		293	265	258	175	140	350	235				18	365	305	170	55	Φ15	100	200	110	20					
150	265	225	8-Φ19	400	285	410	300	210	4-Φ22	485		300		22	390	410	260	275			60	Φ15	100		290	190	20	
200	320	280		400	300	445	350	250	4-Φ25	550		325			440										330	195	20	
250	375	335	12-Φ19	460	360	555	430	325		635		310	24	450	395										215	20		
300	445	400	12-Φ23	550	415	570	410	345		740		400	30	570	480	305	340	65		110	395				225	23	Φ60×3	
350	505	460	16-Φ23	580	420	615	400	350		860		490	33	680	500	320	360		78		Φ20		435	230	25			
400	565	515	16-Φ28	630	490	665	510	430		960		560	36	775	750	660	460					115	480	235	25			
500	670	620	20-Φ28	732	570	752	550	475	4-Φ28	1160		655	38	900				750	660				460	78	Φ20	565	250	26
600	780	725	20-Φ31	830	720	850	700	590	4-Φ33	1320	710	40	1020	830				740	560				660			275	30	

Pump appearance and mobile installation



WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Stirring (JY) H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange	
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7		kg	PN	
1	40WQA10-10-0.75	40	0.75	2950	220/ 380	20	135	200	460	380	110	90	95	100	35	23	6	
2	40WQA8-15-1.1		1.1													24		
3	40WQA15-12-1.1															25		
4	40WQA12-15-1.5		1.5													29		
5	50WQA10-7-0.55	50	0.55				135	200	460	380	110	90	95	100	35	23		
6	50WQA10-10-0.75		0.75													24		
7	50WQA8-15-1.1		1.1													25		
8	50WQA15-12-1.1															29		
9	50WQA15-16-1.5	65	1.5			25	145	230	545	455	120	105	95	105	40	29		
10	65WQA25-8-1.5															38		
11	50WQA15-22-2.2		50				2.2	150	230	580	490	120	105	100		110		37
12	50WQA25-15-2.2																	
13	65WQA30-11-2.2	65							160	230	590	500	125	105		110		115
14	80WQA40-10-2.2	80					170	230	605	515	130	110	110	125		43		
15	100WQA50-7-2.2	100					30	180	230	605	515	130	105	125		140		43
16	50WQA15-30-3	50	3			380	25	150	230	590	500	120	105	110	115	61		
17	50WQA25-20-3		4				170	260	635	535	140	120	110	120	45	68		
18	50WQA15-35-4		5.5													95		
19	50WQA25-28-4		7.5													44		
20	50WQA20-40-5.5	65					155	230	595	505	125	105	110	115	40	62		
21	50WQA25-35-5.5															69		
22	50WQA25-40-7.5															97		
23	65WQA35-15-3		3													44		
24	65WQA35-20-4	80	4				170	260	625	525	140	115	115	120	45	63		
25	65WQA30-30-5.5		5.5													74		
26	65WQA30-35-7.5		7.5													100		
27	80WQA40-13-3		3													64		
28	80WQA50-15-4	100	4				195	320	735	620	160	135	130	135	40	76		
29	80WQA65-17-5.5		5.5													103		
30	80WQA45-25-7.5		7.5													125		
31	100WQA60-9-3		3				205	320	750	640	165	135	135	145	45	140		
32	100WQA60-11-4	150	4													135		
33	100WQA65-15-5.5		5.5													145		
34	100WQA80-17-7.5		7.5													145		
35	100WQA45-25-7.5	150	5.5	1475	40	215	320	765	650	175	135	170	45	103				
36	150WQA100-10-5.5		7.5											125				
37	150WQA100-12-7.5		5.5											140				
38	150WQA130-8-5.5		7.5											135				
39	150WQA150-10-7.5		145															

WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?	
		mm	kw	r/min		V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg		PN
14	200WQA340-7-11	200	11	1475	380	80	420	380	1180	890	245	240	280	310	295	10	yes	
	200WQA400-8-15		15						315									
	200WQA400-10-18.5		18.5						380									
	200WQA400-12-22		22						395									
15	200WQA240-12-11	200	11	1475		85	420	300	1130	830	205	230	270	300	293	6		
	200WQA300-13-15		15						315									
	200WQA300-16-18.5		18.5						380									
	200WQA340-16-22		22						395									
16	200WQA340-22-30	200	30	1475		80	460	400	1560	980	215	250	280	310	655	6	no	
	200WQA400-24-37		37						682									
	200WQA400-28-45		45						738									
	200WQA400-34-55		55						771									
17	200WQA550-32-75	200	75	1475		80	560	400	1900	1230	235	305	340	365	1070	6		
	200WQA600-36-90		90						1145									
18	200WQA480-50-110	200	110	1475		75	550	400	2150	1390	270	325	340	360	1585	10		
	200WQA500-55-132		132						1652									
	200WQA500-64-160		160						1757									
19	250WQA380-7-11	250	11	1475		75	450	460	1140	850	225	250	300	335	305	6	yes	
	250WQA400-10-15		15						325									
	250WQA480-10-18.5		18.5						388									
	250WQA480-12-22		22						405									
20	250WQA600-12-30	250	30	1475		80	530	400	1570	990	210	265	315	360	672	6		
	250WQA600-15-37		37						700									
	250WQA600-18-45		45						755									
	250WQA600-24-55		55						788									
21	250WQA500-13-30	250	30	980		70	560	475	1640	1060	270	290	320	360	690	10	no	
	250WQA600-14-37		37						720									
22	250WQA650-16-45	250	45	980		90	570	550	1970	1300	285	300	335	370	975			
	250WQA700-20-55		55		1024													
23	250WQA700-27-75	250	75	1475	85	520	460	1935	1270	255	280	315	345	1095	6	yes		
	250WQA750-30-90		90					1170										
24	250WQA600-40-110	250	110	980	40	630	650	2280	1530	270	370	420	445	1865	10	no		
25	250WQA650-40-110	250	110	1475	100	660	460	2180	1430	280	325	360	390	1615	6			
	250WQA800-40-132		132					1685										
	250WQA800-50-160		160					1788										

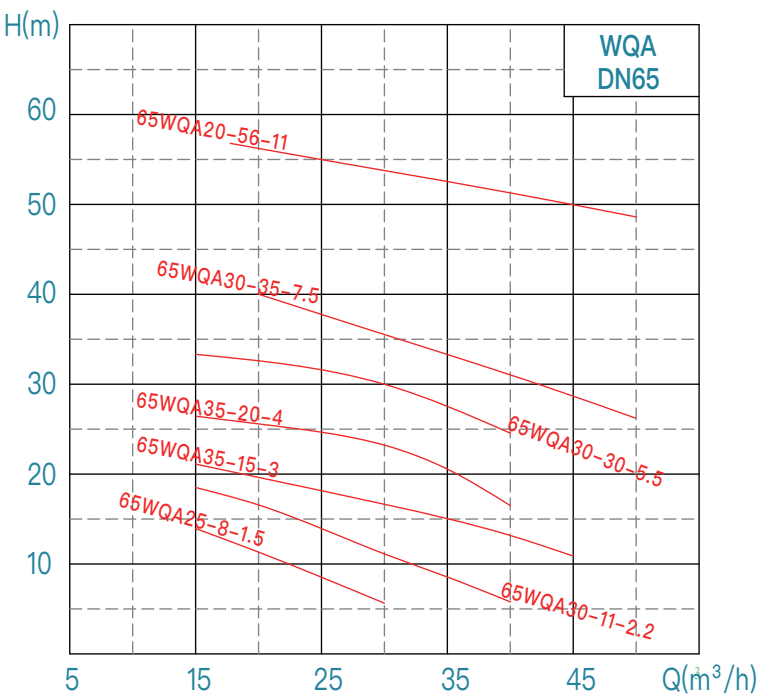
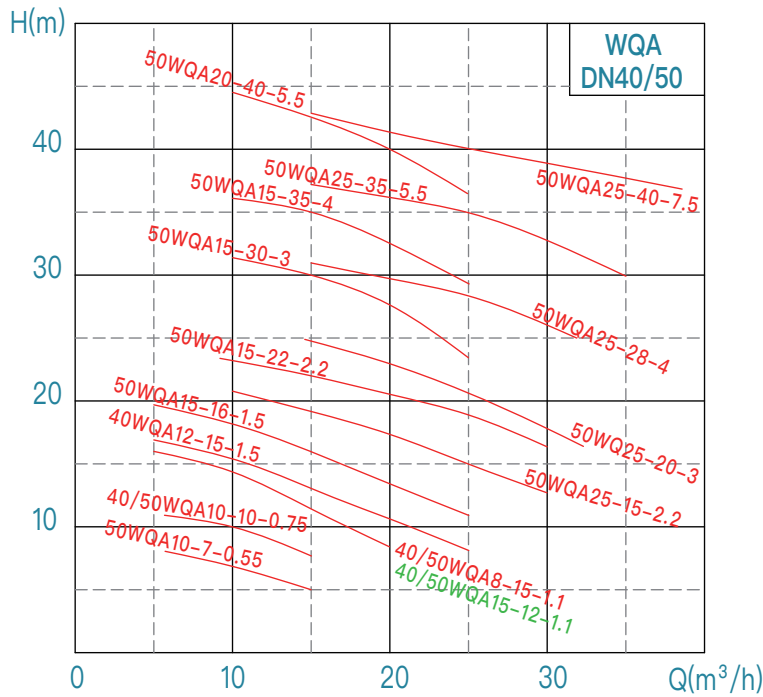
WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?		
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg	PN			
26	250WQA600-68-185	250	185	1475	380	70	600	460	2270	1510	250	350	380	405	1855	10	no		
	250WQA600-74-200		200						2350	1610					1896				
	250WQA600-80-220		220												1963				
	250WQA600-85-250		250												2060				
	250WQA700-90-280		280												2650			1750	2350
27	300WQA600-6-15	300	15	980		90	575	500	1450	950	255	325	380	440	510	10	yes		
	300WQA600-8-18.5		18.5						1500	1000					542				
	300WQA700-8-22		22												558				
28	300WQA800-9-30	300	30	980		90	600	550	1705	1120	285	340	380	425	707	10			
	300WQA800-10-37		37						1785	1200					737				
	300WQA900-12-45		45						1970	1300					983				
	300WQA900-16-55		55												1035				
29	300WQA800-22-75	300	75	980		70	700	650	2180	1420	270	415	475	525	1635	10			
	300WQA900-24-90		90						2300	1540					1721				
	300WQA1000-27-110		110												1891				
30	300WQA900-35-132	300	132	980		45	630	650	2420	1660					1997				
31	300WQA900-20-75	300	75	1475		120	565	525	1985	1320	275	325	390	440	1100	10			
	300WQA900-24-90		90												1175				
32	300WQA1000-25-110	300	110	1475		100	670	525	2205	1450	285	350	400	445	1650	10			
	300WQA1000-30-132		132						2325	1570					1717				
	300WQA1100-36-160		160												1840				
33	300WQA1250-36-185	300	185	1475		70	650	525	2280	1520	265	375	425	470	1910	10		no	
	300WQA1300-38-200		200												1950				
	300WQA1300-42-220		220												1975				
	300WQA1300-50-250		250						2236										
	300WQA1400-52-280		280						2340										
34	350WQA950-8-30	350	30	980		110	740	520	1670	1080	275	410	480	545	792	10			
	350WQA1100-8-37		37						1750	1160					871				
	350WQA1150-10-45		45						1970	1300					1055				
	350WQA1200-12-55		55												1105				
35	350WQA1200-18-90	350	90	980		100	700	650	2180	1420	270	415	475	525	1610	10		yes	
	350WQA1200-20-110		110						2300	1540					1812				
36	350WQA1100-16-75	350	75	980		130	850	625	2210	1450	295	445	515	575	1668	10			
	350WQA1400-15-90		90												1755				
	350WQA1600-17-110		110						2330	1570					1927				
	350WQA1700-20-132		132												2032				

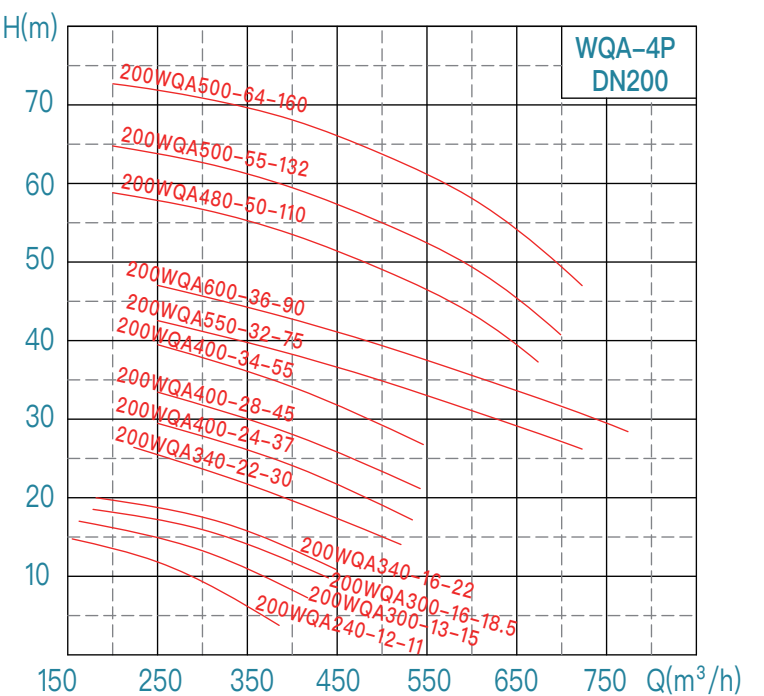
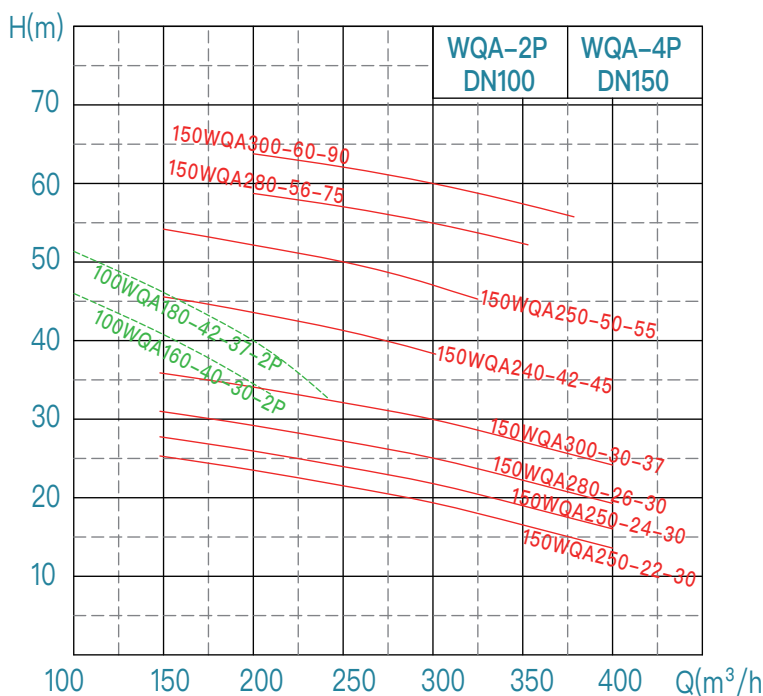
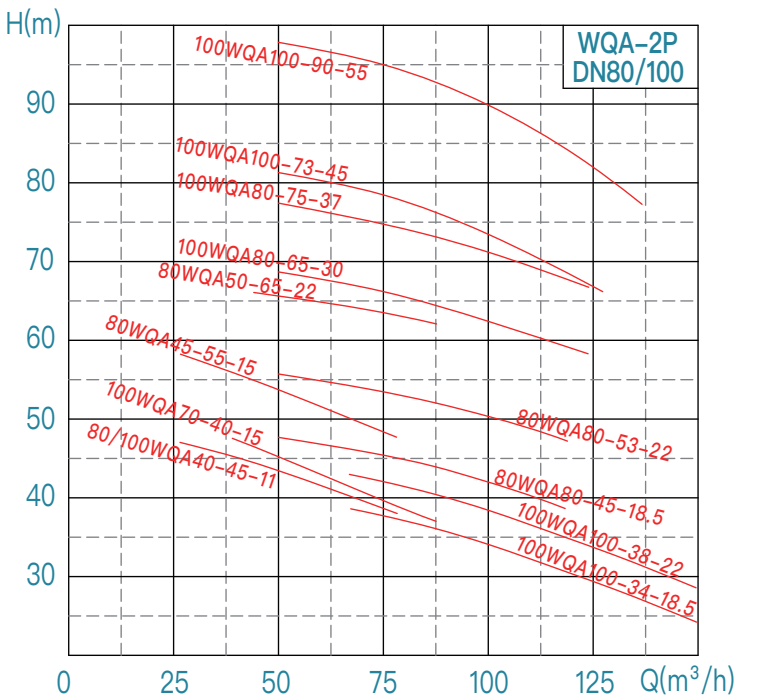
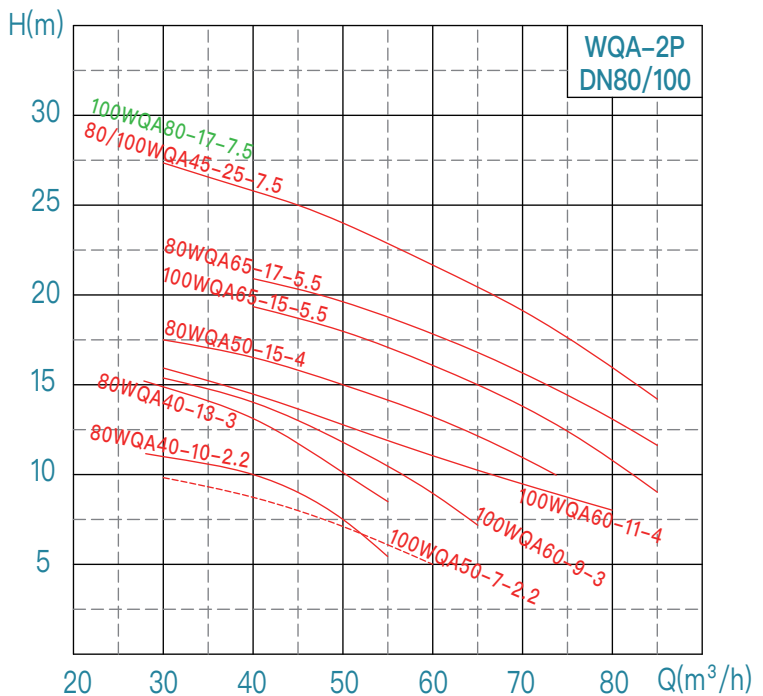
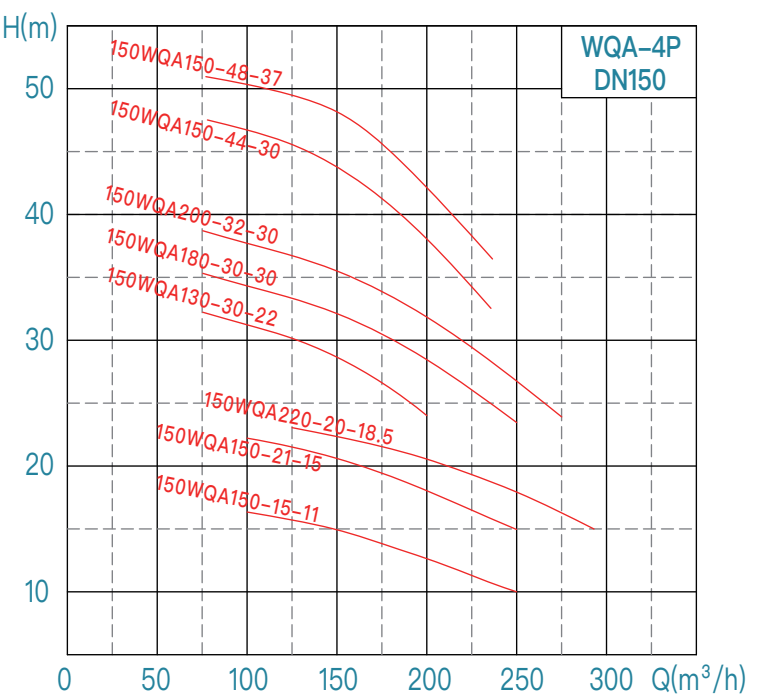
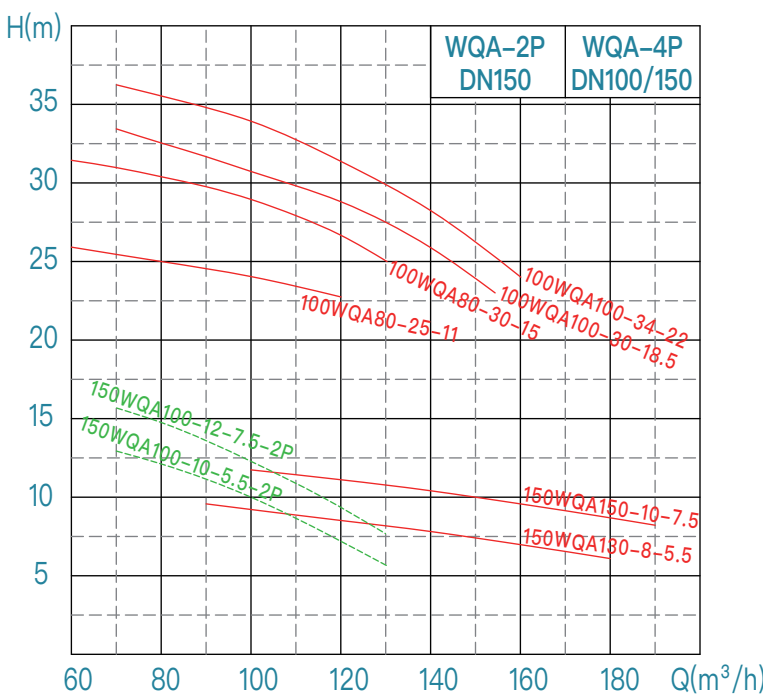
WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg	PN	
37	350WQA1100-35-160	350	160	980	380	55	800	750	2730	1800	355	450	500	540	2570	10	yes
	350WQA1100-40-185		185												2620		
	350WQA1200-40-200		200												2675		
	350WQA1400-40-220		220												2800		
	350WQA1600-40-250		250												2990		
	350WQA1700-42-280		280												3070		
38	400WQA1600-11-75	400	75	980		120	850	650	2280	1520	350	495	565	635	1766	10	
	400WQA1800-12-90		90						2400	1640					1855		
	400WQA1900-15-110		110												2024		
	400WQA2300-14-132		132												2130		
39	400WQA1600-25-160	400	160	980		120	860	650	2750	1820	330	490	560	615	2620	10	
	400WQA1800-28-185		185												2670		
	400WQA1800-30-200		200												2730		
40	400WQA1900-26-220	400	220	980		150	850	670	2800	1870	335	470	555	635	2850	10	
	400WQA2200-28-250		250												2995		
	400WQA2200-32-280		280												3070		
41	400WQA1200-6-30	400	30	740		100	850	660	1960	1290	350	500	580	650	1700	10	
	400WQA1300-7-37		37						2060	1390					1737		
	400WQA1400-8-45		45												1816		
	400WQA1500-10-55		55												1910		
42	400WQA2000-9-75	400	75	740		100	800	800	2400	1640	435	475	575	655	2050	10	
	400WQA2200-10-90		90						2520	1760					2350		
	400WQA2200-12-110		110						2910	1990					2680		
	400WQA2200-15-132		132		2735												
	400WQA2400-17-160	160	2910	1990	3000												
	400WQA2500-18-185	185			3200												
	400WQA2500-19-200	200			3400												
43	500WQA2500-8-90	500	90	740	100	800	800	2520	1760	435	475	575	655	2330	10		
	500WQA2500-10-110		110											2390			
	500WQA3000-10-132		132											2780			
	500WQA3000-14-160		160					3130									
	500WQA3000-16-185		185					3210									
	500WQA3000-18-200		200					3290									
44	500WQA3000-20-220	500	220	740	95	1000	800	2770	1840	365	620	710	790	3470	10		
	500WQA3000-23-250		250					2950	2020					3550			
	500WQA3000-25-280		280											3740			
45	600WQA3500-12-185	600	185	740	150	960	870	2950	2020	460	570	690	785	3380	10		
	600WQA4000-12-200		200											3460			
	600WQA4000-14-220		220											3640			
	600WQA4000-16-250		250					3720									
	600WQA4200-17-280		280					3910									
	600WQA4200-20-315		315					4090									

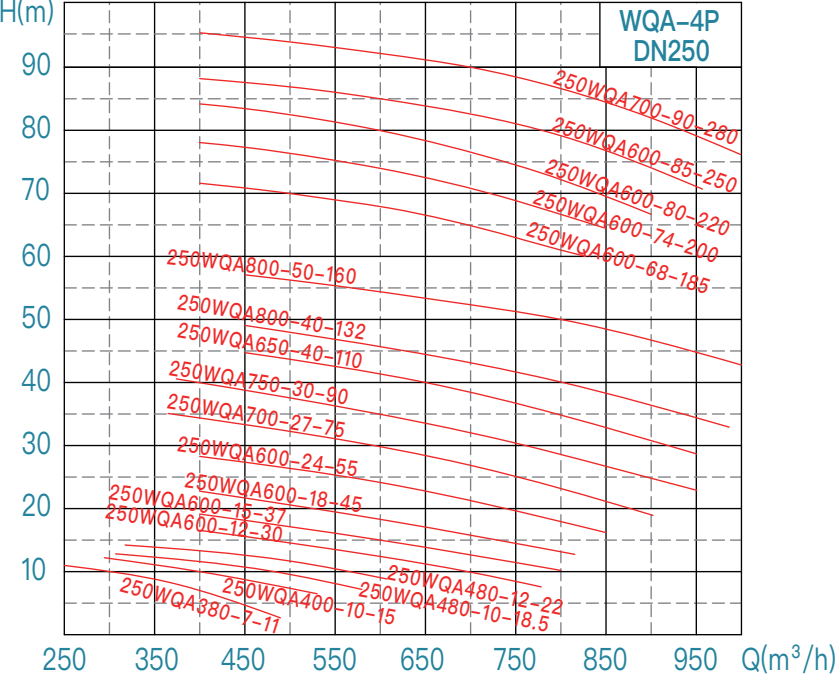
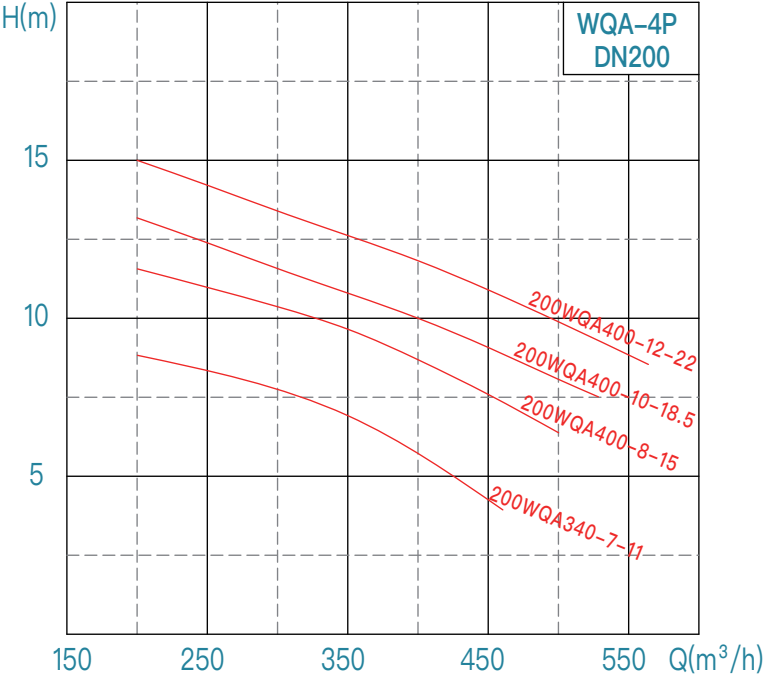
WQA performance curve chart (50Hz)



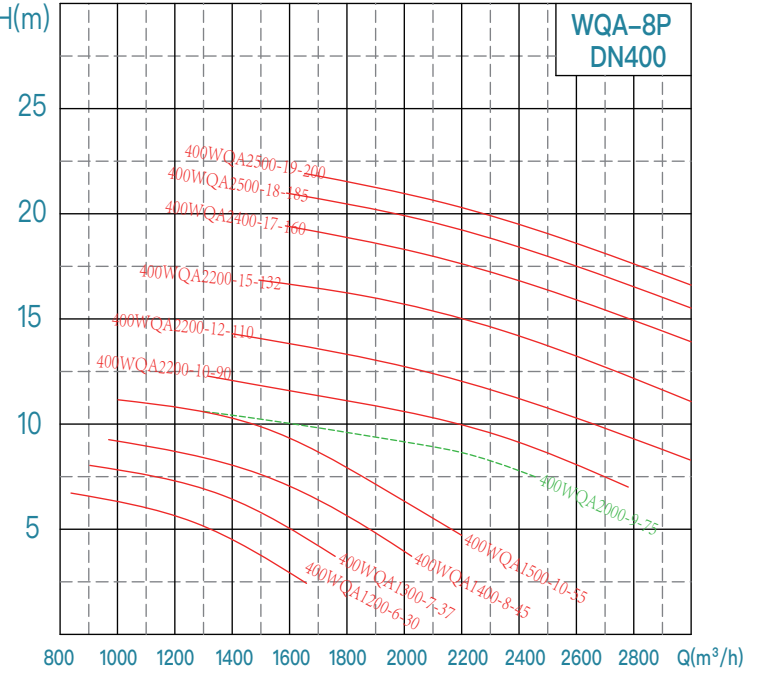
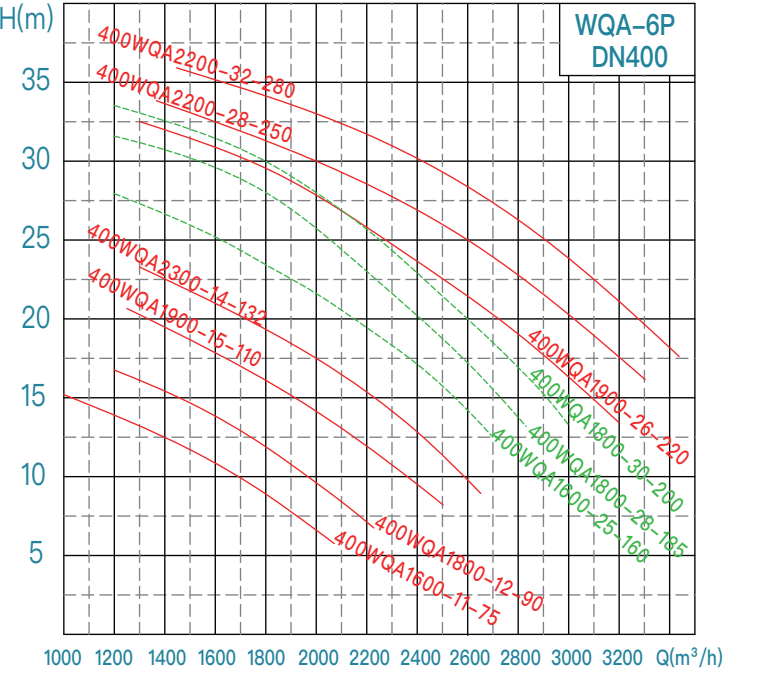
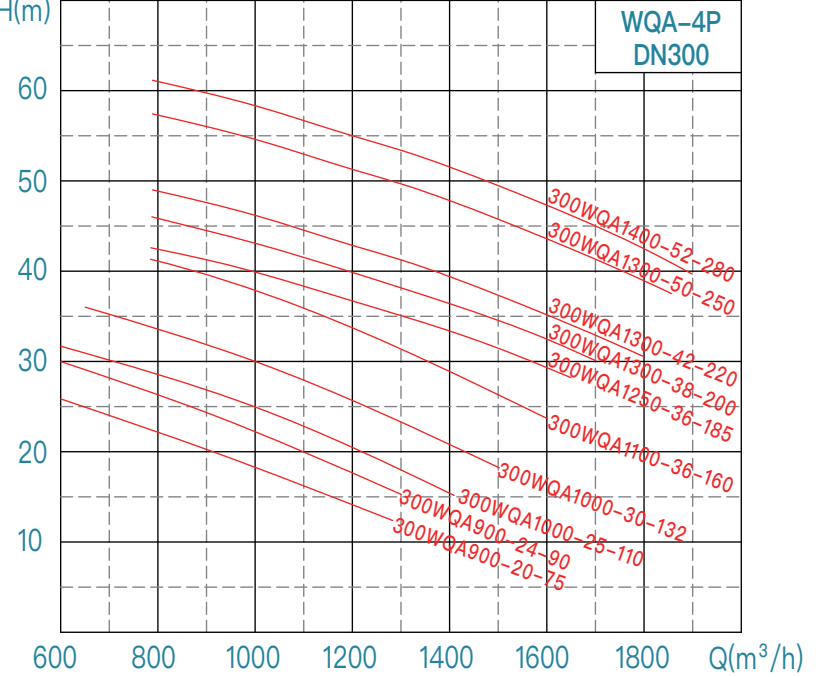
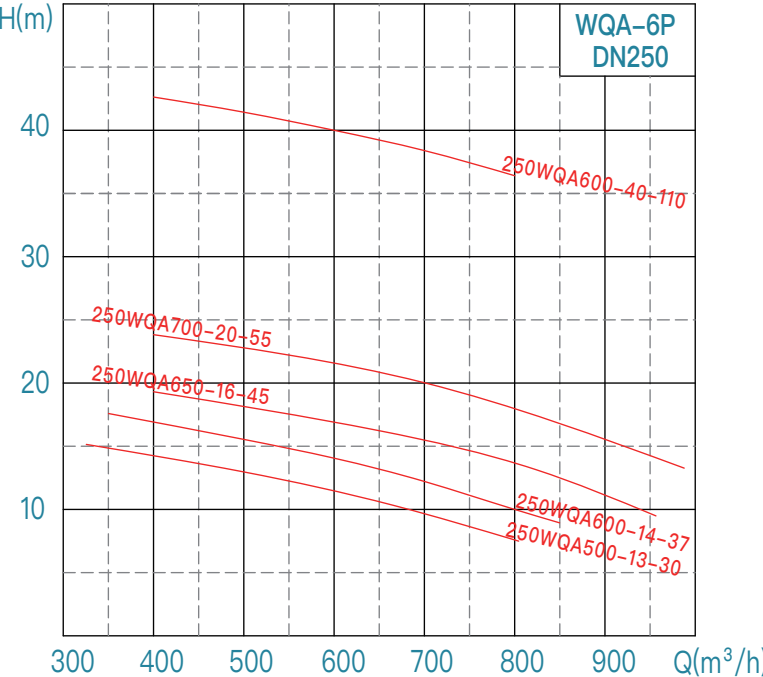
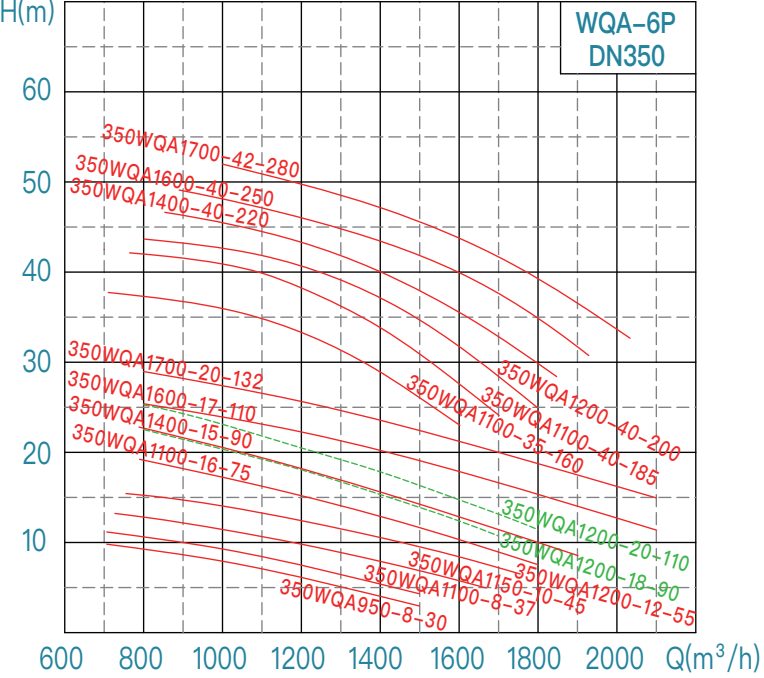
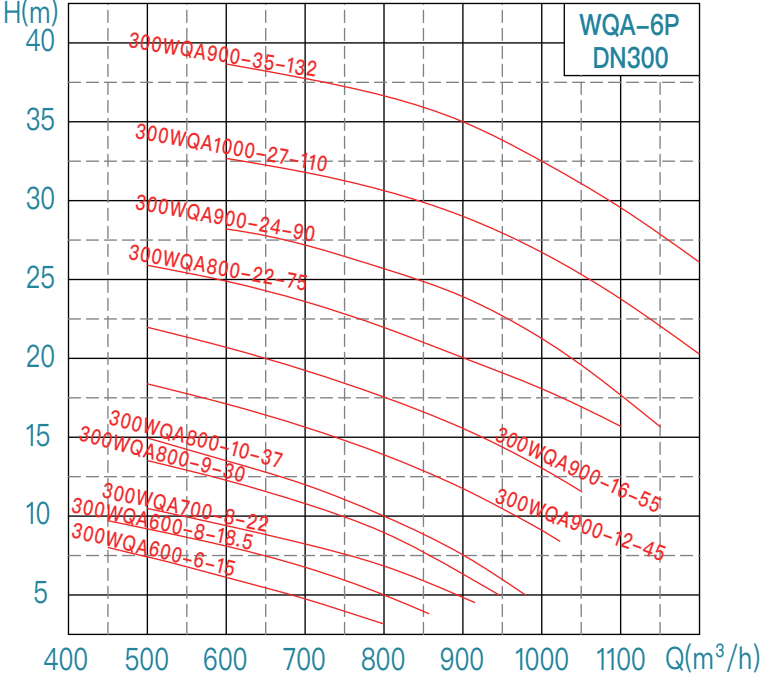
WQA performance curve chart (50Hz)



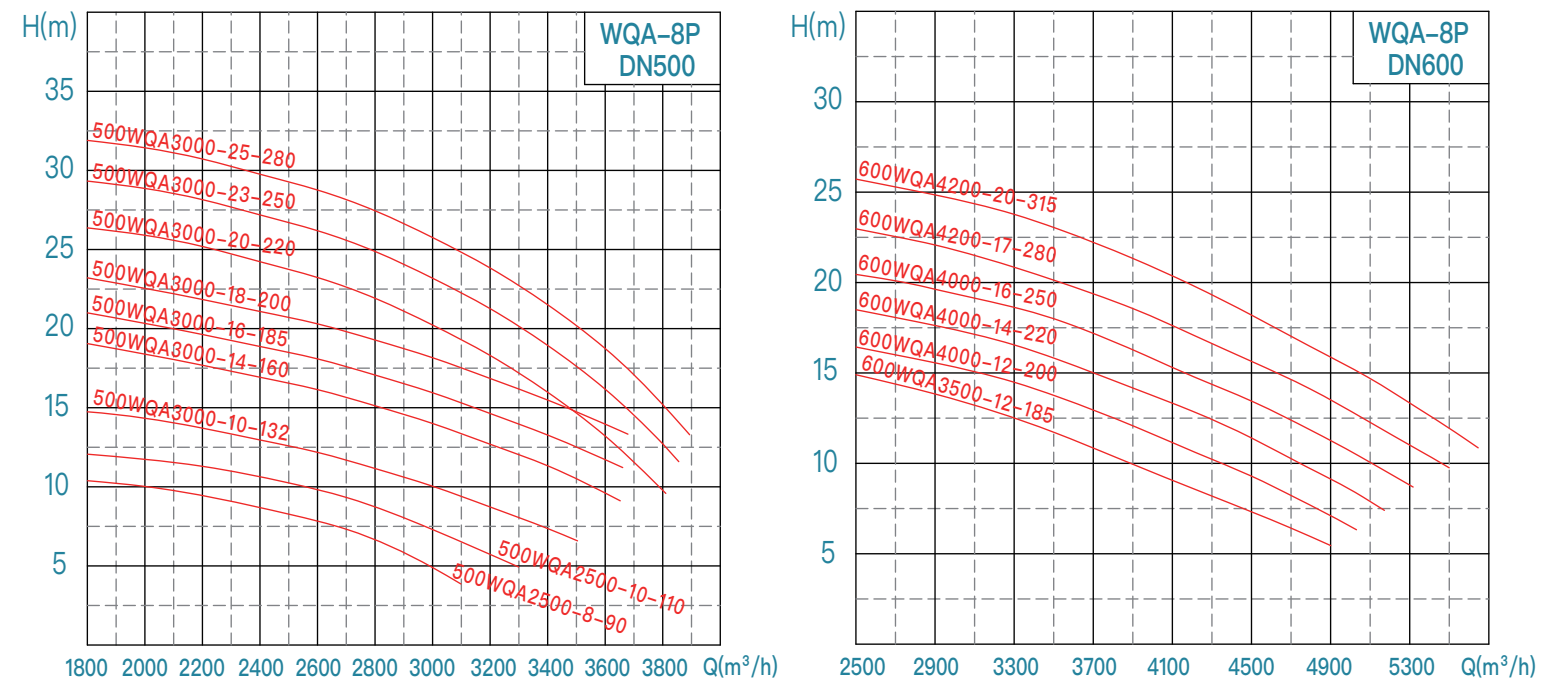
WQA performance curve chart (50Hz)



WQA performance curve chart (50Hz)



WQA performance curve chart (50Hz)



WQE economic
submersible sewage pump ►

WQE

Economic

Submersible sewage pump



Scope of Application:
municipal engineering, sewage treatment, flood control and drainage, hydraulic engineering, industry sewage, environmental protection ...

Operating Conditions:
Flow Rate: 7 ~ 2500 m³/h
Head: 7 ~ 75 m
Medium Temperature: T ≤ 40°C
Medium Density: ≤ 1100 kg/m³; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 10
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

Product Introduction:
The WQE series submersible sewage pumps are designed to provide better reliability and higher efficiency for our customers. Based on over 30 years of experience in submersible pump manufacturing and technological accumulation, we have optimized the hydraulic model. Additionally, we have comprehensively optimized the mechanical structure, sealing, and protection aspects, and adopted high-quality motor materials to ensure the stability of the unit. The WQE series consists of a total of 116 models.



Structure diagram of WQE products

Main materials and configuration table

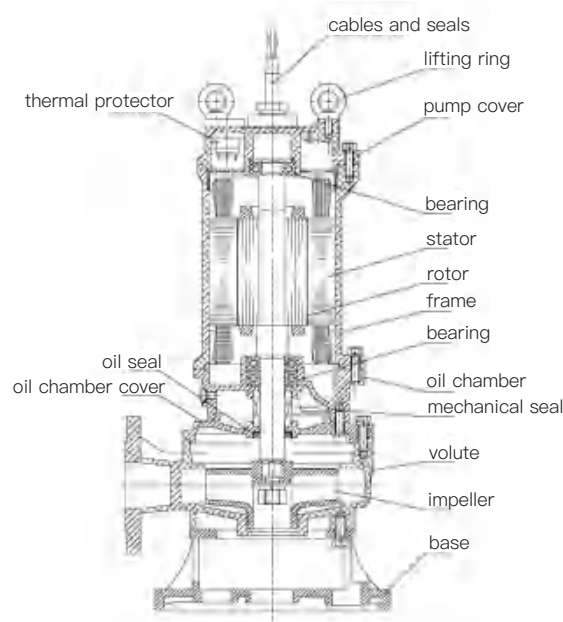
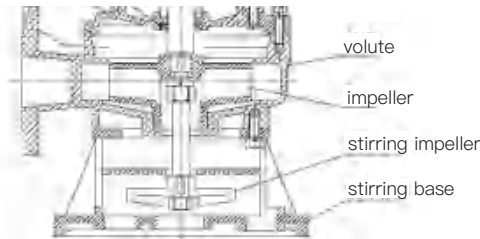


Figure 1

Structure diagram for 7.5kW and below



structure diagram for stirring base

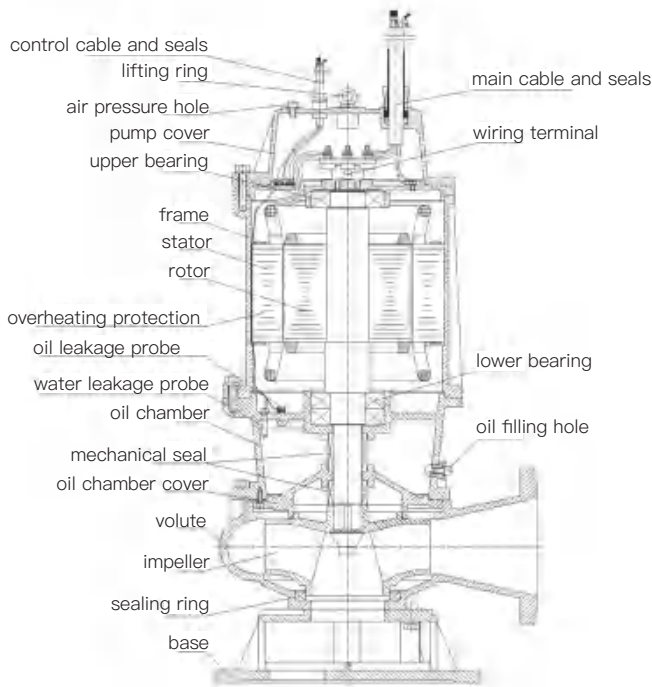


Figure 2

Structure diagram for 11kW and above

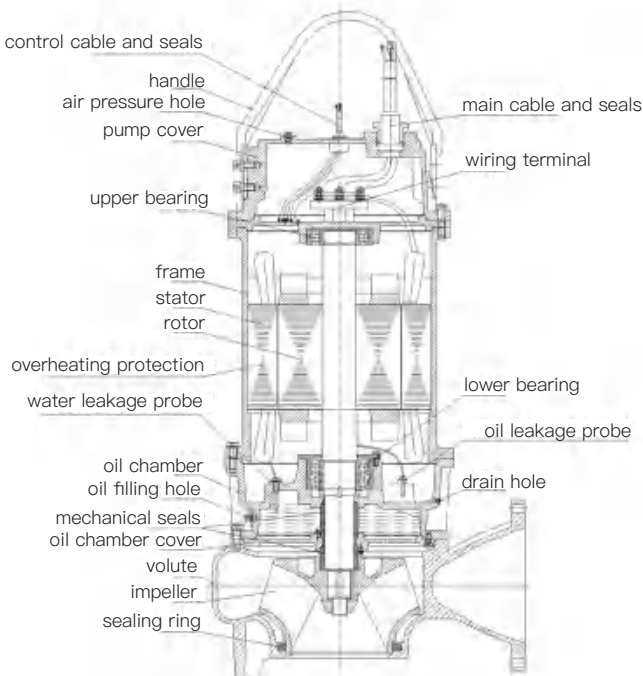


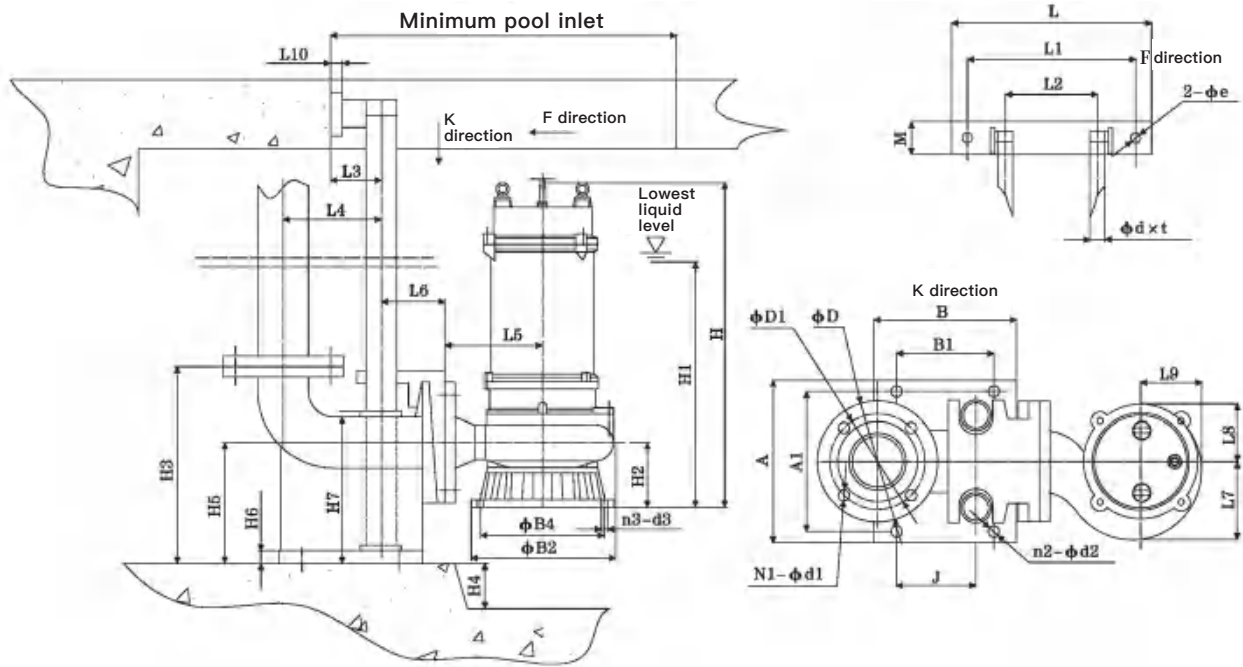
Figure 3

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		HT200	/
2	Impeller		HT200	/
3	Oil chamber cover		HT200	/
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		HT200	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		2Cr13	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	/
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/SiC (Q1Q2)	SiC/WC (Q1U1)
11		Spring	Stainless steel	/
12		Rubber parts	(NBR)	/
13	O-ring		(NBR)	/
14	External leakage fastener		7.5kW and below: 8.8 11kW and above: A2-70	7.5kW and below: A2-70
15	Protection device (standard)	75-6P, 90-6P 110kW-132kW	Water leakage probe/Float switch /Overheating protection	Thermal protector for 3kW and below

Specification/model of submersible motor cable

Motor power (kW)	Cable model	Standard main cable specification	Star delta	Model of controlling cable
0.55	YZW 300/500V	YZW 4×0.75	/	YZW 4×0.75 (Non–standard)
0.75				
1.1		YZW 4×1		
1.5				
2.2		YZW 3×1.5+1×1		
3				
4		YZW 3×2.5+1×1.5		
5.5				
7.5		YZW 3×2.5+1×4		
11/15–2P				
11/15–4P	2–YZW 3×2.5+1×1.5	YZW 4×1 (Non–standard)		
18.5	2–YZW 3×6+1×4			
22				
30	2–YCW 3×10+1×6			
37	YCW 3×16+1×16			
45	YCW 3×25+1×10			
55	YCW 3×50+1×16		2–YCW 3×25+1×10	
75–4P			2–YCW 3×35+1×10	
90–4P			2–YCW 3×25+1×10	
75–6P			2–YCW 3×35+1×10	
90–6P	2–YCW 3×35+1×10		YZW 3×1.5+1×1 (Standard)	
110	2–YCW 3×35+1×10			
132	2–YCW 3×50+1×16			

installation dimension drawing



Flange coupling parts dimension table

Model	Flange Connection Size			Coupling Base Size																						
DN	D	D1	N1-Φd1	A	A1	B	B1	J	n2-Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φd×t		
40	130	100	4-Φ14	108	65	138	68	68	4-Φ15	195	Adjust according to pump model	120	14	150	225	185	70	35	Φ14	65	125	85	13	Φ33×3		
50	140	110		197	168	208	130	120	4-Φ19	250		165		/	265	215	105	42	Φ14	66	135	90	15			
65	160	130		238	190	252	155	135		265		170	16		280	235	125	50	Φ14	70	155	85	18			
80	190	150	4-Φ19	255	220	225	153	125	310	190		365			305	145	53	Φ14	80	155	100	18	Φ48×3			
100	210	170		293	265	258	175	140	350	235			18		365	305	170	55	Φ15	100	200	110		20		
150	265	225	8-Φ19	400	285	410	300	210	4-Φ22	485		300	22	390	410	260	275	60			290	190		20		
200	320	280		400	300	445	350	250	4-Φ25	550		325		440							330	195		20		
250	375	335	12-Φ19	460	360	555	430	325		635		310	24	450							395	215		20		
300	445	400	12-Φ23	550	415	570	410	345	4-Φ25	740		400	30	570	480	305	340	65		110	395	225	23	Φ60×3		
350	505	460	16-Φ23	580	420	615	400	350		860		490	33	680	500	320	360			480	235	25				
400	565	515	16-Φ28	630	490	665	510	430		960		560	36	775	750	660	460									
500	670	620	20-Φ28	732	570	752	550	475	4-Φ28	1160		655	38	900			750		660	460	78	Φ20		565	250	26
600	780	725	20-Φ31	830	720	850	700	590	4-Φ33	1320		710	40	1020			830	740	560							

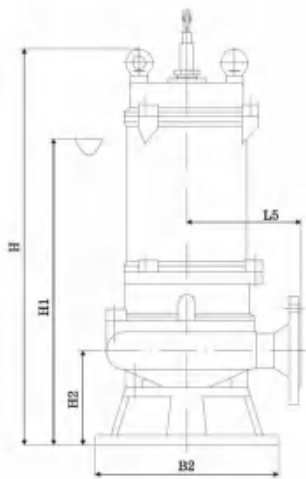
Lightweight flange coupling parts dimension table

Model DN	Flange Connection Size			Coupling Base Size						H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φd×t
	D	D1	N1–Φd1	A	A1	B	B1	J	n2–Φd2															
50	140	110	4–Φ14	142	120	170	120	120	4–Φ14	235	100	150	11	180	225	185	72	36	Φ11	60	133	53	12	Φ33×3
65	160	130		150	128	195	142	142		245		138	12	177	265	215					142	55		
80	190	150	4–Φ18	170	148	210	158	158		272		158	13	220	280	235	77			77	153	82	Φ48×3	
100	205	170		190	158	220	145	143	4–Φ18	340		198	14	257	315	265					178			
150	260	225	8–Φ18	262	220	306	230	240	4–Φ22	425	150	268	18	350	365	305	95	52	Φ15	83	230	100	18	

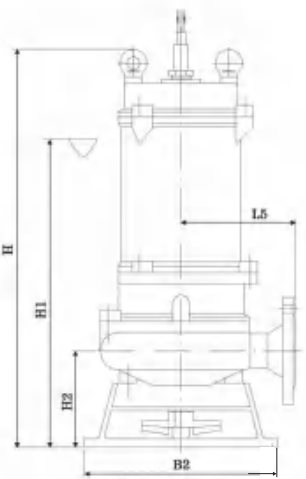
WQE installtion dimension table

No.	WQE model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)										304 impeller	Stirring (JY) H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange						
		mm	kW	r/min	V	mm	L5	B2	B4	n3-d3	H	H1	H2	L8	L9	L7			kg	kg						
1	40WQE10-10-0.75	40	0.75	2950	380	10	125	175	/	/	520	370	90	90	95	100	/	30	22	6						
2	50WQE10-7-0.55	50	0.55																22							
3	50WQE10-10-0.75		0.75				22																			
4	40WQE7-15-1.1	50	1.1				155	215			600	430	125	95	100	115			26							
5	50WQE7-15-1.1					22																				
6	50WQE15-15-1.5		1.5			15	150	235			570	440	125	105	110	125			22							
7	50WQE15-20-2.2																		22							
8	65WQE25-10-1.5	1.5	580								450	620	460	640	480	135			105		110	135	22			
9	65WQE25-15-2.2																						27			
10	80WQE40-10-2.2	80	2.2			30	165	640			480	135	105	110	135	27										
11	100WQE50-7-2.2	100				35	170	650			490	135	105	115	135	34										
12	50WQE15-27-3	50	3			25	150	640			480	130	115	125	135	34										
13	50WQE15-31-4		4													700			540		145	125	130	150	35	
14	50WQE15-37-5.5		5.5			30	720	560			37															
15	50WQE20-40-7.5		7.5			25	210	330			680	600	150	145	155	170			60							
16	65WQE30-15-3	65	3			30	165	235			650	500	125	120	135	145		30	40							
17	65WQE30-23-4		4																780		600	155	130	135	140	40
18	65WQE30-29-5.5		5.5																680		600	150	145	155	170	39
19	65WQE30-35-7.5		7.5															40								
20	80WQE40-12-3	80	3			25	180	235			660	500	135	110	135	140		30	43							
21	80WQE40-16-4		4																700		460	155	130	140	150	57
22	80WQE40-22-5.5		5.5																720		580					57
23	80WQE45-25-7.5		7.5															720	640		160	140	145	155	60	58
24	100WQE60-9-3	100	3			35	180	235			670	500	130	120	135	140		30	58							
25	100WQE6-10-4		4																740		560	160	110	125	140	59
26	100WQE65-15-5.5		5.5																760		580					64
27	100WQE80-15-7.5		7.5															720	640		160	140	145	155	60	64
28	150WQE100-9-5.5	150	5.5			45	210	270			780	600	175	125	155	160		30	64							
29	150WQE100-12-7.5		7.5																760		680	180	145	160	180	65
30	150WQE135-10-7.5		7.5															850	720		230	170	185	205	65	
31	65WQE20-56-11	65	11	2950	380	25	215	330			850	720	200	145	155	165	●	60	119							
32	80WQE40-45-11	80	11								850	720	220	145	155	165			120							
33	80WQE45-55-15		15								900	770	220	145	155	165			130							
34	100WQE80-25-11	100	11			35	330	410	357	4-Φ14	920	730	210	220	235	255			220							
35	100WQE80-30-15		15								950	750							240							
36	100WQE100-25-15										950	750							240							
37	100WQE80-35-18.5		18.5								1050	850						280								
38	100WQE100-30-18.5										1050	850							280							
39	100WQE100-37-22		22								1150	950						300								

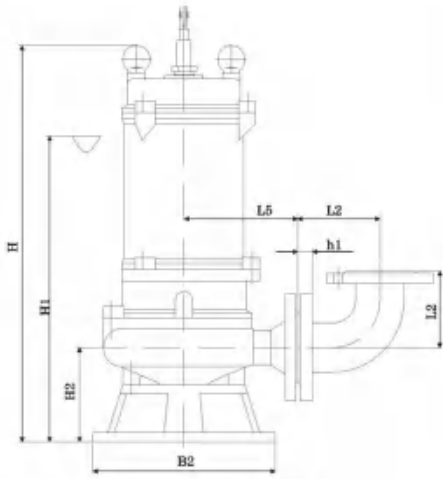
Pump appearance and moblie installtion



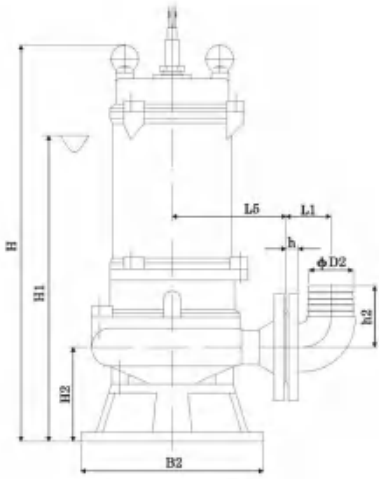
Dimension drawing of pump appearance



Dimension drawing of stirring pump (JY)



Hard pipe (double flange elbow/joint) mobile installation



Soft pipe (elbow) mobile installation

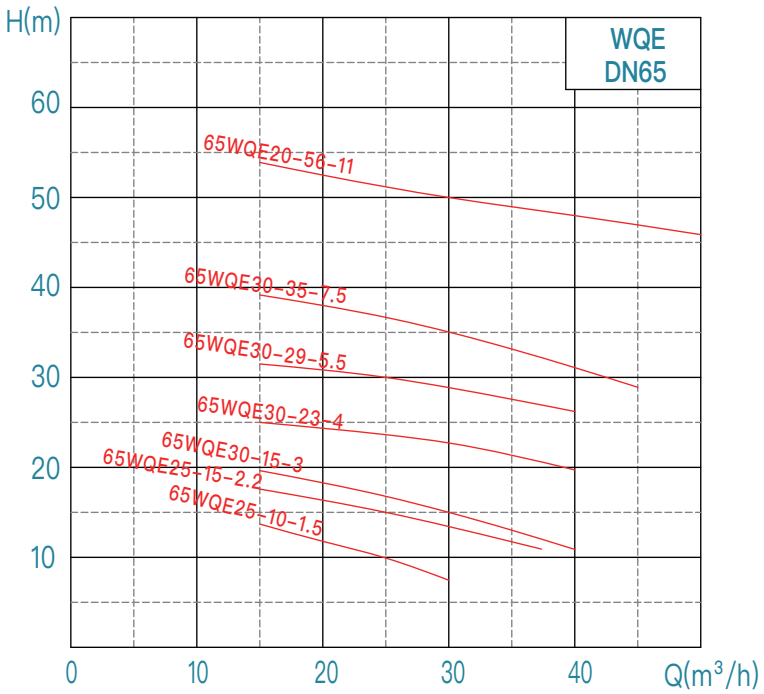
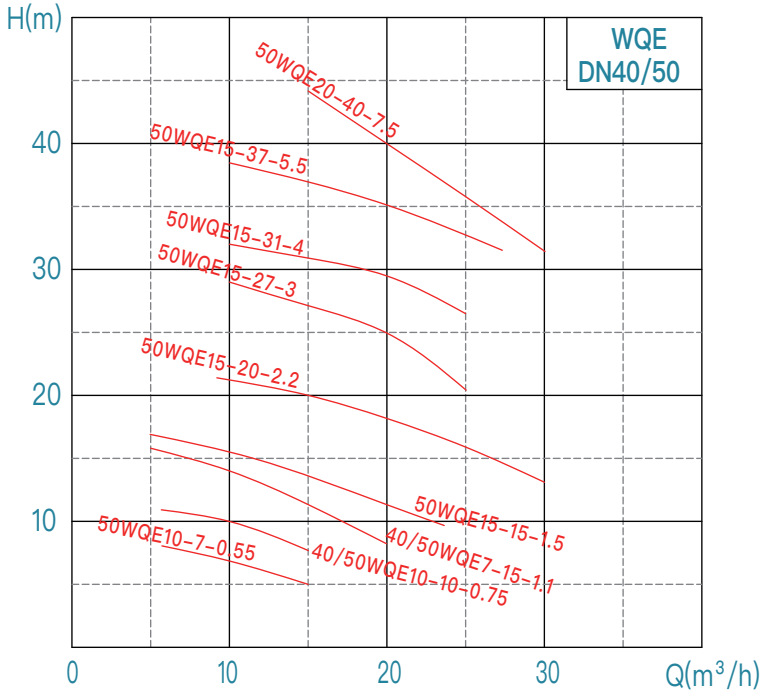
WQE installtion dimension table

No.	WQE model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)										304 impeller	Stirring (JY) H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange
		mm	kW	r/min	V	mm	L5	B2	B4	n3-d3	H	H1	H2	L8	L9	L7			kg	kg
40	150WQE130-15-11	150	11	1475	380	45	355	430	380	4-Φ14	950	750	260	205	225	255	●	60	220	6
41	150WQE170-18-15		15				355	430	380	4-Φ14	1000	800	260	205	225	255		60	240	
42	150WQE180-20-18.5		18.5				355	430	380	4-Φ14	1100	900	260	205	225	255		80	290	
43	150WQE180-25-22		22				355	430	380	4-Φ14	1150	950	260	205	225	255		80	300	
44	200WQE300-7-11	200	11			55	365	430	380	4-Φ14	950	750	260	215	235	265	●	60	230	6
45	200WQE250-13-15		15				365	430	380	4-Φ14	1000	800	260	215	235	265		60	250	
46	200WQE250-15-18.5		18.5				365	430	380	4-Φ14	1100	900	260	215	235	265		80	290	
47	200WQE250-17-22		22				365	430	380	4-Φ14	1150	950	260	215	235	265		80	310	
48	250WQE400-6-11 *	250	11			80	350	400	340	4-Φ22	1000	800	245	210	260	310	/	/	255	10
49	250WQE400-8-15 *		15				350	400	340	4-Φ22	1100	850	245	210	260	310			275	
50	250WQE400-9-18.5 *		18.5				370	400	340	4-Φ22	1120	910	235	235	285	325			315	
51	250WQE400-10-22 *		22				370	400	340	4-Φ22	1120	910	235	235	285	325			335	
52	100WQE120-42-30 *	100	30			35	360	515	435	4-Φ28	1380	970	285	260	265	270	/	/	435	6
53	100WQE130-45-37 *		37				360	515	435	4-Φ28	1500	1050	285	260	265	270			510	
54	100WQE120-55-45 *		45				360	515	435	4-Φ28	1600	1150	285	260	265	270			520	
55	100WQE150-60-55 *		55				360	515	435	4-Φ28	1650	1200	285	260	265	270			650	
56	150WQE180-30-30	150	30			45	380	520	467	4-Φ16	1250	1000	290	250	280	310	●	80	435	6
	150WQE250-22-30						380	520	467	4-Φ16	1250	1000	290	250	280	310		80	435	
57	150WQE180-34-37		37				380	520	467	4-Φ16	1250	1000	290	250	280	310		0	500	
	150WQE250-27-37						380	520	467	4-Φ16	1250	1000	290	250	280	310		0	500	
58	150WQE250-34-45	200	45			55	380	520	467	4-Φ16	1300	1050	270	250	280	310	/	0	510	6
59	150WQE200-50-55 *		55				410	515	435	4-Φ28	1600	1150	245	265	285	310		/	640	
60	200WQE360-16-30		30			55	390	520	467	4-Φ16	1250	1000	290	250	280	310	●	80	445	
61	200WQE350-22-37		37				390	520	467	4-Φ16	1250	1000	290	250	280	310		0	520	
62	200WQE400-25-45	250	45			70	390	520	467	4-Φ16	1350	1100	300	270	300	330	/	0	530	10
63	200WQE300-38-55 *		55				400	515	435	4-Φ28	1650	1200	265	265	270	290		/	670	
64	250WQE600-9-30		30				390	520	467	4-Φ16	1300	1050	300	270	300	340		80	455	
65	250WQE600-13-37	300	37			90	390	520	467	4-Φ16	1350	1100	300	270	300	340	●	0	530	
66	250WQE600-17-45		45				390	520	467	4-Φ16	1400	1150	300	270	300	340		0	540	
67	250WQE600-20-55 *		55				440	515	435	4-Φ28	1700	1250	285	275	320	360		/	710	
68	300WQE800-7-30 *		30			40	410	490	/	/	1400	1000	255	270	320	370			480	
69	300WQE900-7-37 *	37	410				490	1500			1050	255	270	320	370	550				
70	300WQE900-9-45 *	45	410				490	1550			1100	255	270	320	370	560				
71	300WQE800-15-55 *	55	410				490	1700			1250	255	270	320	370	730				
72	350WQE1000-6-30 *	350	30	980	1475	90	520	590	/	/	1550	1100	280	345	410	460	/	/	505	6
73	350WQE1000-7-37 *		37				520	590			1550	1100	280	345	410	460			656	
74	350WQE1000-10-45 *		45				510	530			1600	1150	285	300	380	450			580	
75	350WQE1100-10-55 *		55				510	530			1700	1250	285	300	380	450			750	
76	150WQE200-55-75 *	150	75	1475	380	40	440	515	435	4-Φ28	1700	1200	255	295	315	335	/	/	720	6
77	150WQE240-60-90 *		90				440	515	435	4-Φ28	1700	1200	255	295	315	335			880	

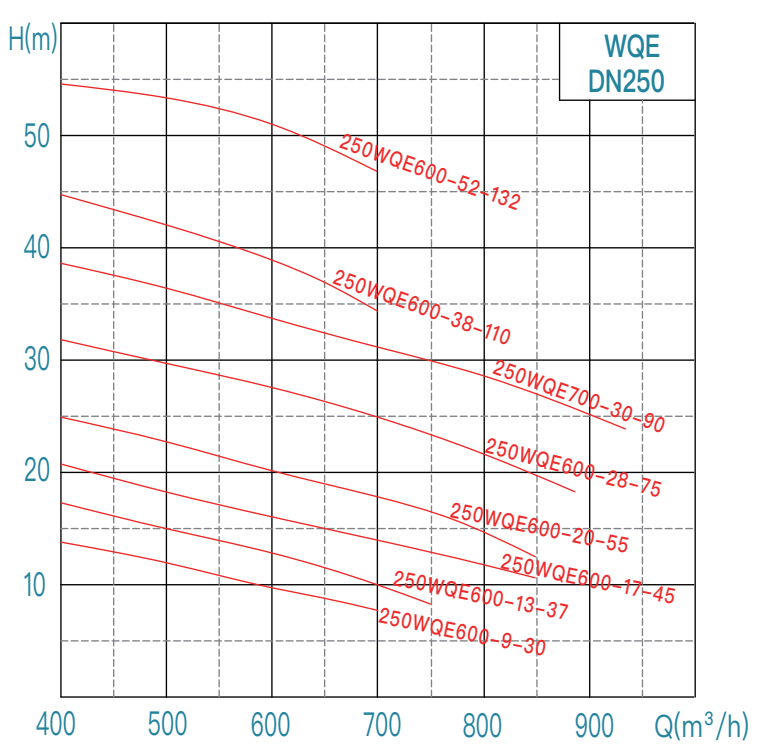
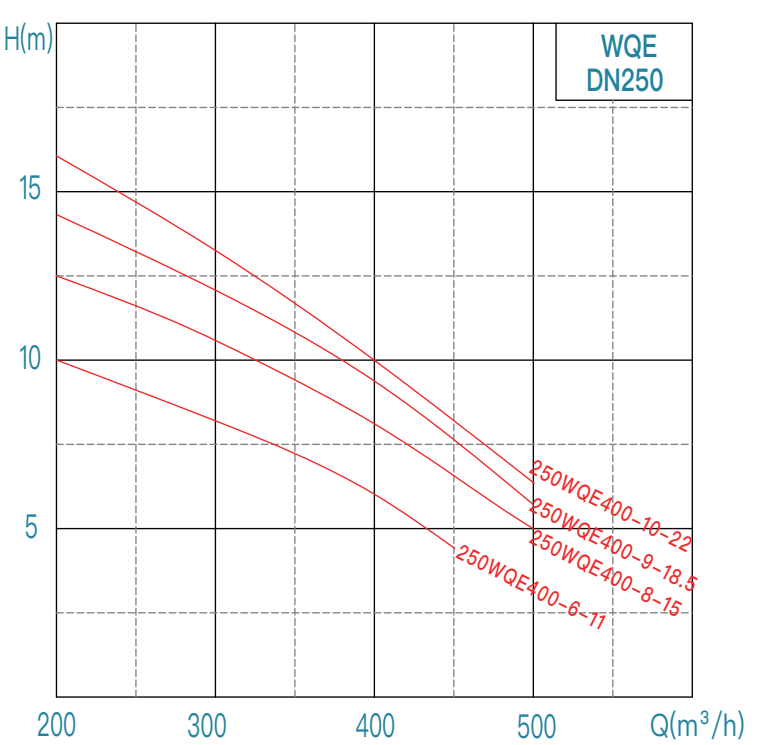
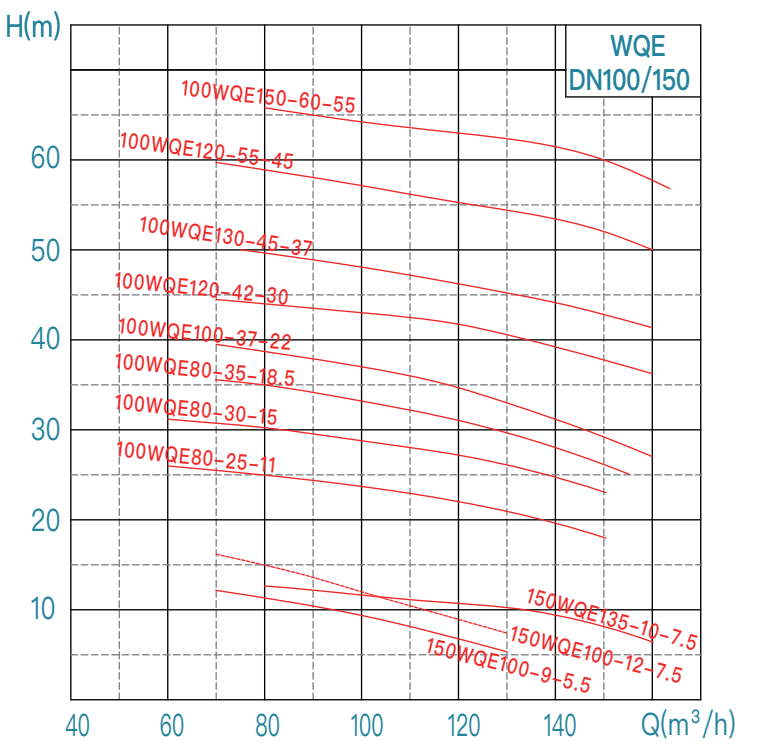
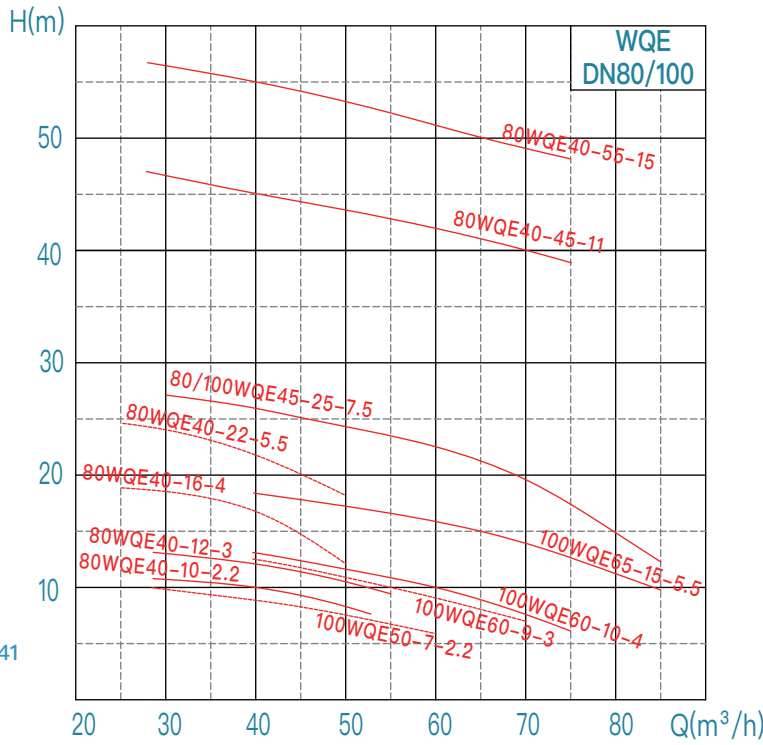
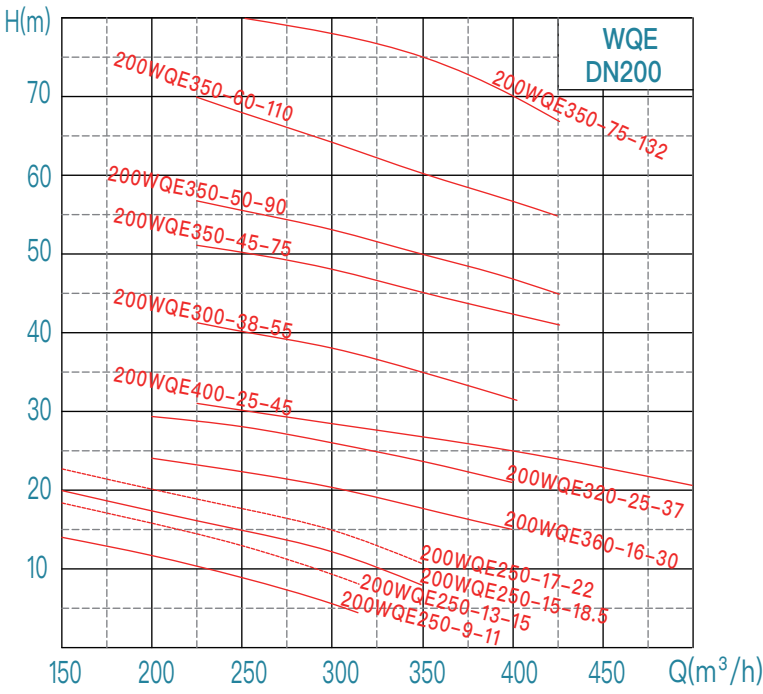
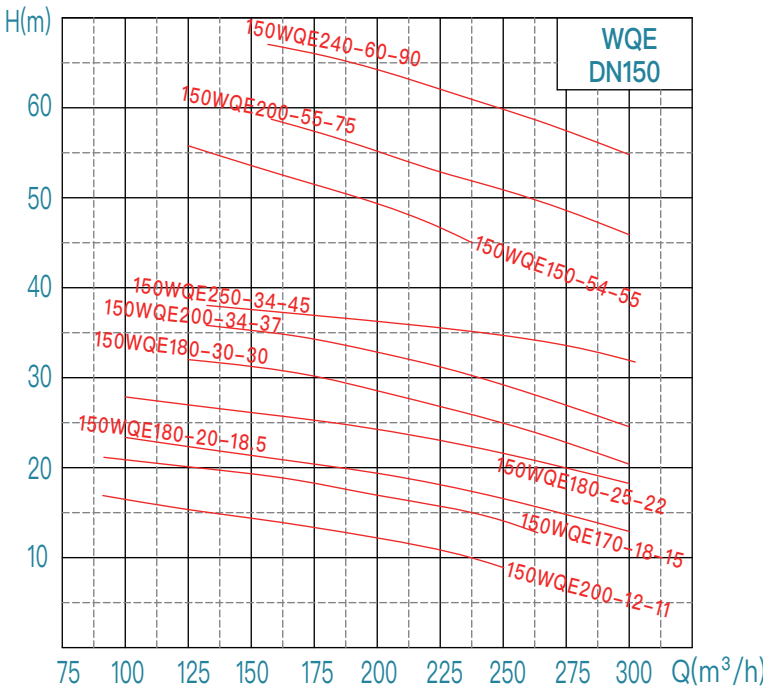
WQE installtion dimension table

No.	WQE model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)										304 impeller	Stirring (JY) H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange	
		mm	kW	r/min	V	mm	L5	B2	B4	n3-d3	H	H1	H2	L8	L9	L7			kg	kg	
78	200WQE350-45-75 *	200	75	1475	380	50	460	515	435	4-Φ28	1750	1250	270	300	325	355	/	/	770	6	
79	200WQE350-50-90 *		90				460	515	435	4-Φ28	1850	1350	270	300	325	355			930		
80	200WQE350-60-110 *		110				545	580	/		1900	1400	215	340	360	385			1300		
81	200WQE350-75-132 *		132				545	580			2000	1500	215	340	360	385			1390		
82	250WQE600-28-75 *	250	75			60	480	515	435	4-Φ28	1800	1300	305	310	350	385			760	10	
83	250WQE700-30-90 *		90				480	515	435	4-Φ28	1900	1400	305	310	350	385			810		
84	250WQE600-38-110 *		110				540	730	660	4-Φ24	2000	1500	320	370	375	380			1300		
85	250WQE600-52-132 *		132				540	730	660	4-Φ24	2100	1600	320	370	375	380			1390		
86	300WQE800-18-75 *	300	75			75	460	515	/			1800	1300	255	280	320			360		830
87	300WQE900-22-90 *		90				460	515				1900	1400	255	280	320			360		980
88	300WQE900-27-110 *		110				485	580				2000	1500	255	385	400			420		1300
89	300WQE800-35-132 *		132				485	580				2200	1600	255	385	400			420		1420
90	350WQE1200-15-75 *	350	75	980	1475	110	565	650	/			2000	1400	285	410	455	510	1165			
91	350WQE1100-18-90 *		90				565	650				2000	1400	285	410	455	510	1440			
92	350WQE1000-24-110 *		110				90	565	650	2000	1500	285	410	455	510	1370					
93	350WQE1400-17-110 *		110				980	110	565	650	2100	1500	285	410	455	510	1640				
94	350WQE1000-30-132 *	400	132			1475	90	565	650	/			2000	1500	285	410	455	510	1450		
95	350WQE1500-18-132 *	400	132	980	120	710	730	660	4-Φ24	2350	1800	420	470	550	620	1650					
96	400WQE1300-8-55 *	400	55	980	380	110	590	640	/			1800	1300	310	380	455	510	926			
97	400WQE1400-12-75 *		75				590	650				2100	1500	315	410	455	510	1180			
98	400WQE1500-15-90 *		90				590	650				2100	1500	315	410	455	510	1470			
99	400WQE1500-17-110 *		110				590	650				2150	1550	315	410	455	510	1670			
100	400WQE2000-13-132 *	400	132			120	715	830	/			2250	1700	315	485	530	620	1680			
101	500WQE1800-7-55 *	500	55			110	690	800	/			1800	1300	365	425	520	600	985			
102	500WQE1800-9-75 *		75				690	800				2000	1450	365	425	520	600	1230			
103	500WQE2000-10-90 *		90				690	800				2000	1450	365	425	520	600	1500			
104	500WQE2500-10-110 *		110				690	800				2200	1650	365	425	520	600	1720			
105	500WQE2500-12-132 *		132				720	830				2300	1750	370	485	555	620	1740			

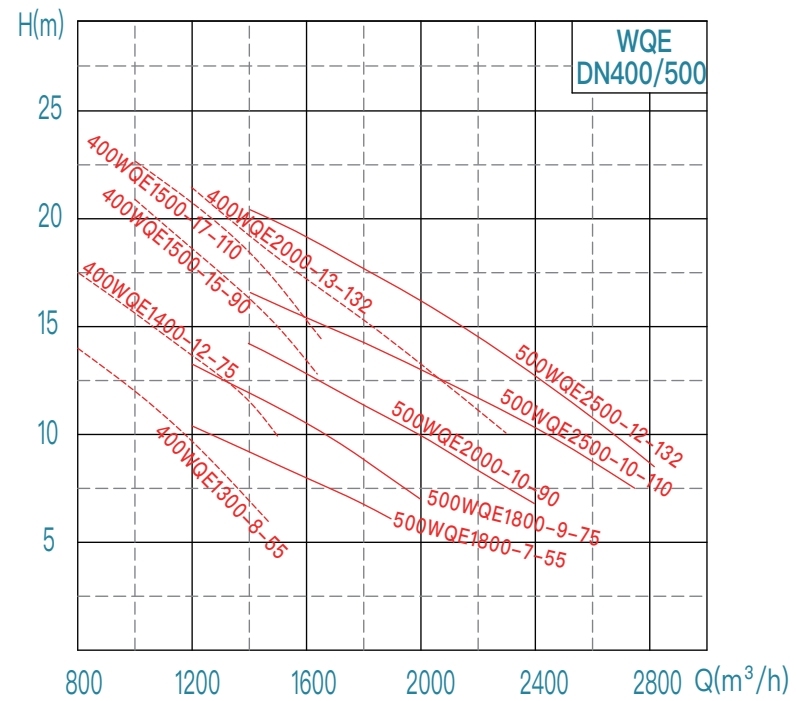
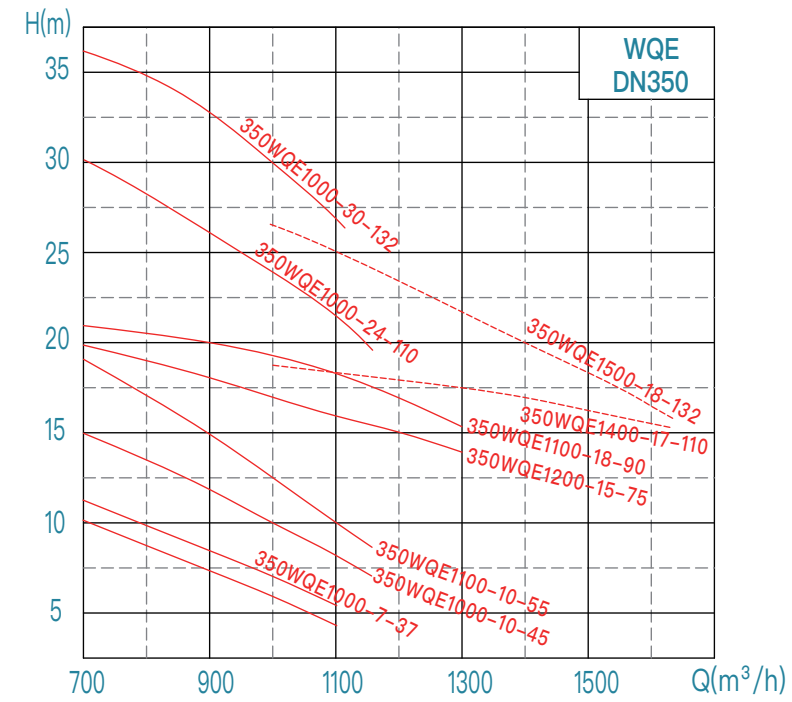
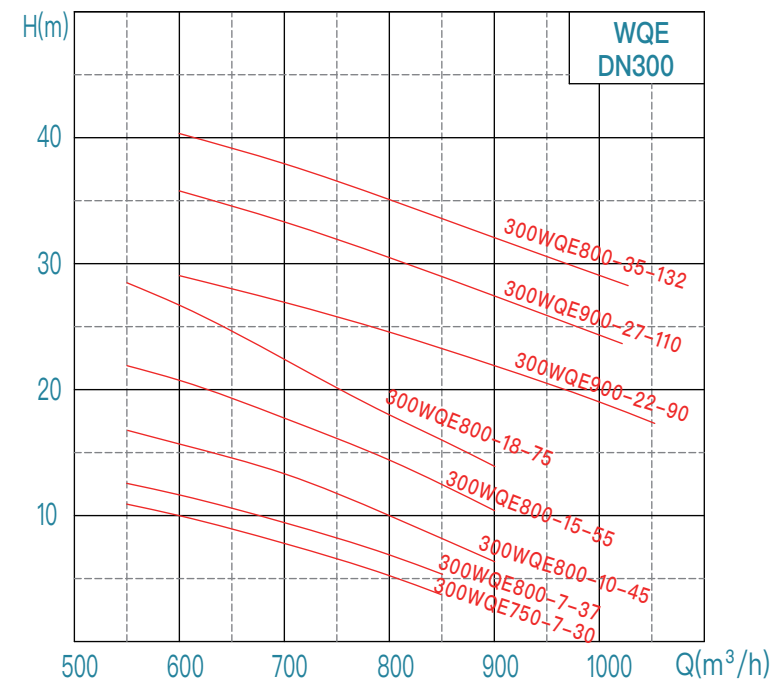
WQE performance curve chart (50Hz)



WQE performance curve chart (50Hz)



WQE performance curve chart (50Hz)



WQF stainless steel
submersible sewage pump ►

WQF

Stainless steel

Submersible sewage pump

Customizable Materials:

304/ 316/ 316L/

2205/ 2507



Scope of application:

municipal engineering,
sewage treatment,
chemical engineering,
hydraulic engineering,
seawater treatment,
environment treatment ...

Operating Conditions:

Flow Rate: 7 ~ 600 m³/h
Head: 7 ~ 55 m
Medium Temperature: T ≤ 40°C
Medium Density: ≤ 1100 kg/m³; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 11
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

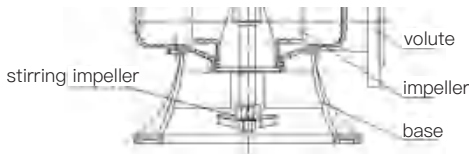
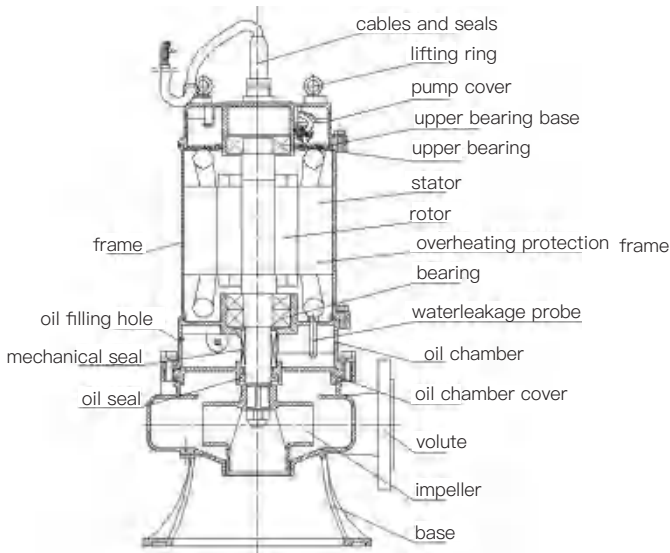
Product Introduction:

The WQF series stainless steel submersible sewage pumps from Hangzhou Xizi Pump Industry are specialized products developed with advanced technology. The pump's flow components are all made of stainless steel through precision casting, ensuring pure quality and an attractive appearance. They are self-pollution-free and corrosion-resistant, expanding the application fields of water supply and drainage.



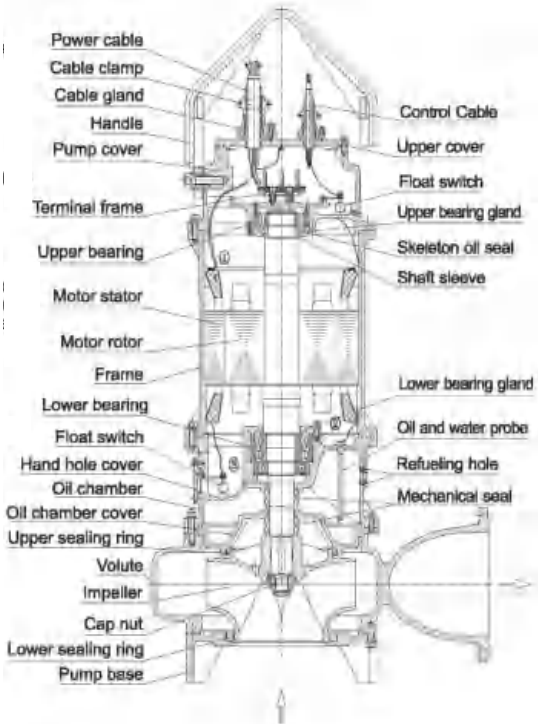
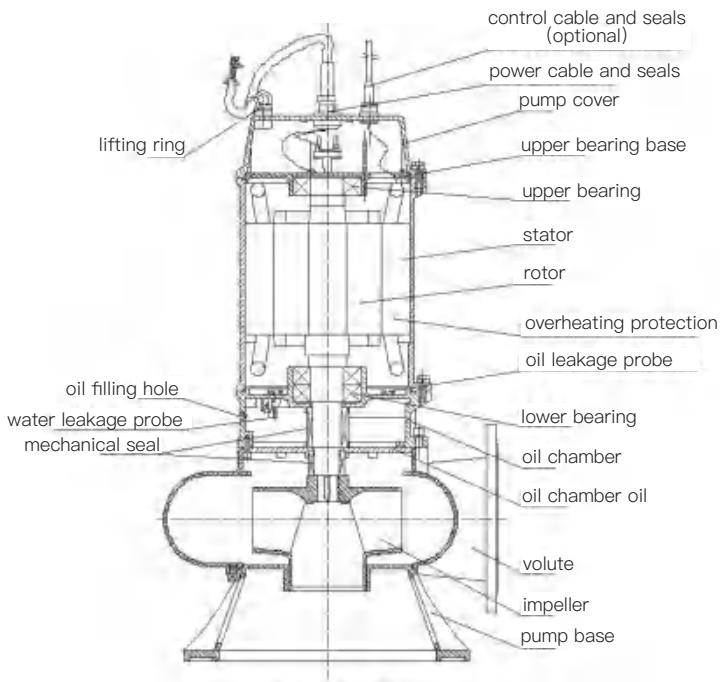
Structure diagram of WQF products

Main materials and configuration table



structure diagram for stirring base

Structure diagram for 7.5kW and below



Structure diagram for 55kW and above

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		304	/
2	Impeller		304	/
3	Oil chamber cover		304	/
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		304	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		304	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/WC (Q1U1)	WC/WC (U1U2)
11		Spring	Stainless steel	/
12		Rubber parts	(P)	/
13	O-ring		(P)	/
14	External leakage fastener		A2-70	/
15	Protection device (optional)	45kW and below	Water leakage probe/Float switch /Overheating protection	Order remarks according to customer requirements.

Installation dimension drawing

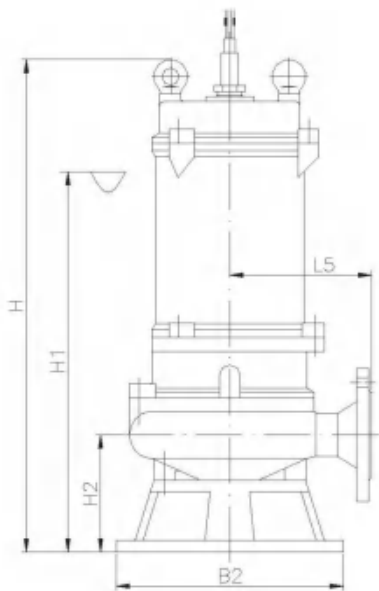
The technical drawing illustrates a vertical pump assembly with three main views:

- Side View (Left):** Shows the pump's vertical profile. Key dimensions include:
 - $L10$: Minimum pool inlet distance.
 - $L3$: Distance from the pool inlet to the pump base.
 - $L4$: Distance from the pump base to the motor base.
 - $L6$: Motor base width.
 - $L5$: Motor base depth.
 - $H1$: Total height from the base to the pump outlet.
 - $H2$: Motor base height.
 - $H3$: Height from the base to the pump inlet.
 - $H5$: Height from the base to the pump inlet.
 - $H6$: Height from the base to the pump inlet.
 - $H7$: Height from the base to the pump inlet.
 - $H4$: Height from the base to the pump inlet.
 - $\Phi B2$: Base diameter.
 - Lowest liquid level**: Indicated by a dashed line.
 - F direction**: Flow direction arrow.
- Front View (Right):** Shows the pump's front profile. Key dimensions include:
 - L : Total length.
 - $L1$: Distance from the front to the pump inlet.
 - $L2$: Distance from the pump inlet to the motor base.
 - M : Motor base width.
 - $2-\Phi e$: Two mounting holes.
 - $\Phi d \times t$: Motor base dimensions.
- Top View (Bottom):** Shows the pump's top profile. Key dimensions include:
 - ΦD : Total diameter.
 - B : Total width.
 - $\Phi D1$: Inner diameter.
 - $B1$: Inner width.
 - A : Total height.
 - $A1$: Inner height.
 - $N1-\Phi d1$: Four mounting holes.
 - $J1$: Distance from the front to the pump inlet.
 - J : Distance from the pump inlet to the motor base.
 - $n2-\Phi d2$: Four mounting holes.
 - $L9$: Distance from the front to the pump inlet.
 - $L8$: Distance from the pump inlet to the motor base.
 - $L7$: Distance from the motor base to the pump inlet.
 - $L6$: Motor base width.
 - $L5$: Motor base depth.
 - $L4$: Distance from the pump base to the motor base.
 - $L3$: Distance from the pool inlet to the pump base.
 - $L10$: Minimum pool inlet distance.
 - K direction**: Flow direction arrow.

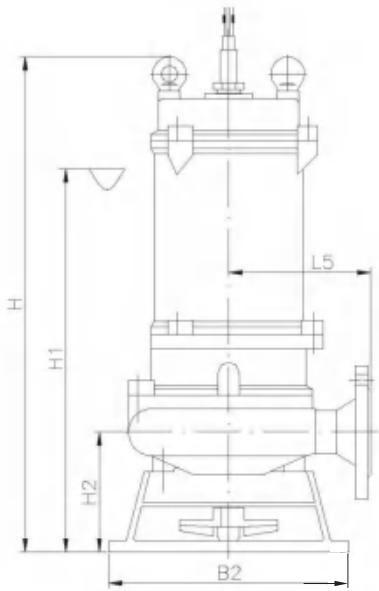
Model				Flange Connection Size				Coupling Base Size																				
DN	D	D1	N1-Φd1	A	A1	B	B1	J	J1	n2-Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φdxt			
50	165	125	4-Φ19	180	140	270	190	214	30	4-Φ19	240	100	120	12	200	220	180	60	44	Φ12	50	152	52	10	Φ33×3			
65	180	145		190	150	280	190	221	30		240	100	120	12	200	220	180	60	44		50	149	55	10				
80	195	160	8-Φ19	210	170	300	210	237	30		270	150	130	12	225	250	210	80	50	Φ15	57	165	63	10	Φ48×3			
100	215	180		230	190	300	210	237	30		300	150	150	14	260	250	210	80	50		57	185	63	10				
150	280	240	8-Φ23	300	250	460	330	367	50		380	150	180	18	320	330	280	120	60		62	305	76	12				
200	340	295		350	300	470	350	370	50		460	200	220	20	390	330	280	120	60		62	320	80	12				
250	395	350	12-Φ23	420	360	580	450	471	50		500	200	240	20	450	340	280	160	70		Φ19	115	359	95		16	Φ76×3.5	
300	445	400		460	400	630	500	521	50		560	200	260	20	490	380	320	200	70			115	399	96		16		
350	505	460	16-Φ23	520	460	680	550	570	50	640	300	300	20	550	380	320	200	70	115	439		88	16					

WQF installtion dimension table

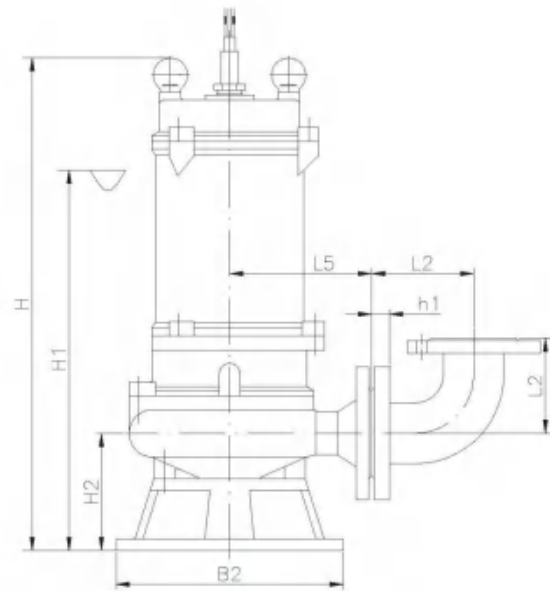
Pump appearance and moblie installtion



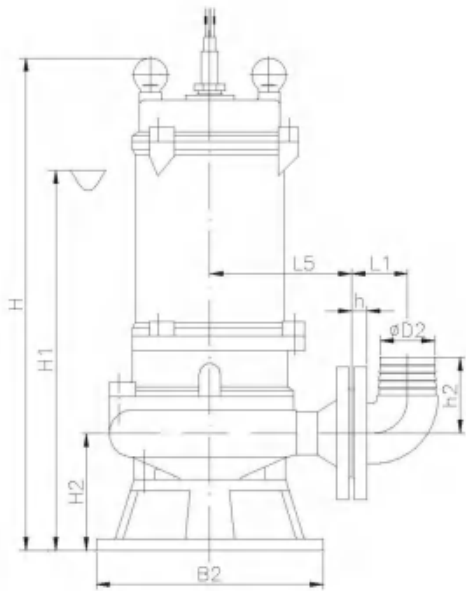
Dimension drawing of pump appearance



Dimension drawing of stirring pump (JY)



Hard pipe (double flange elbow/joint) mobile installtion



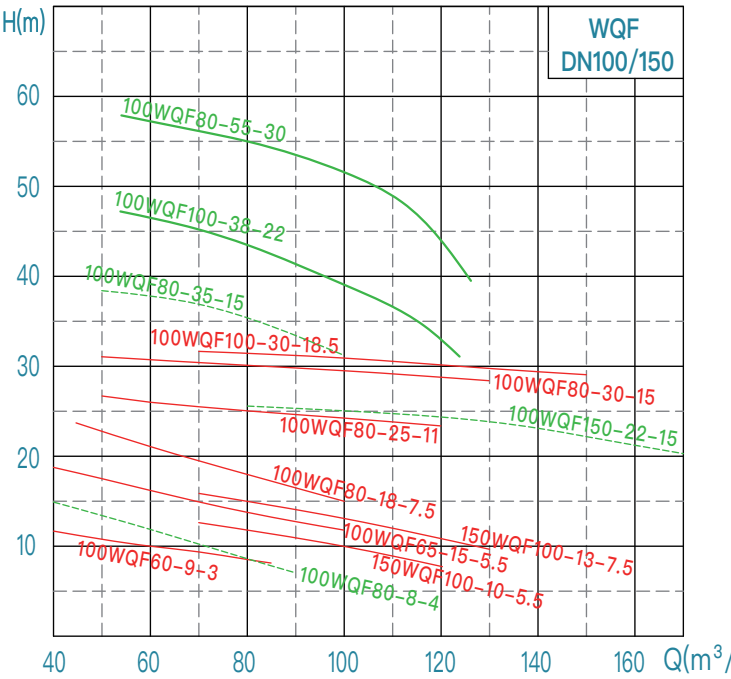
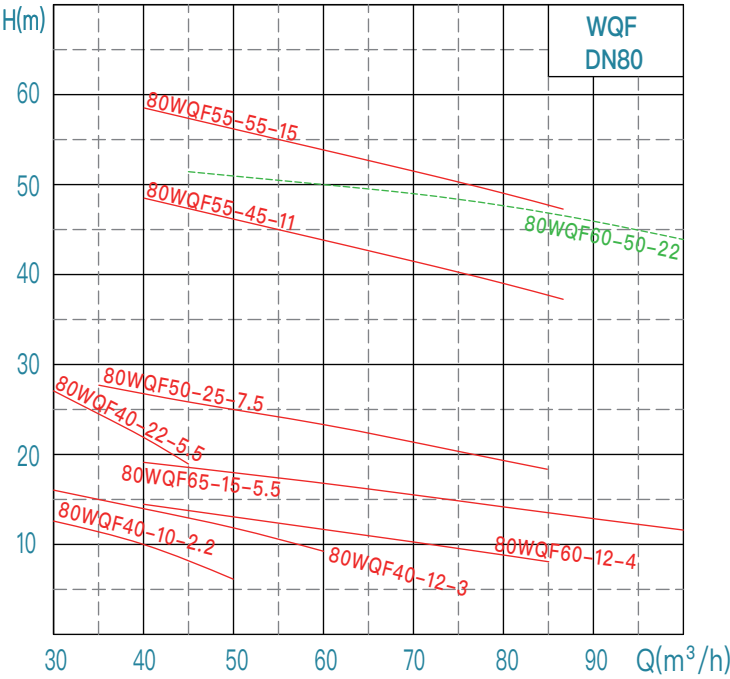
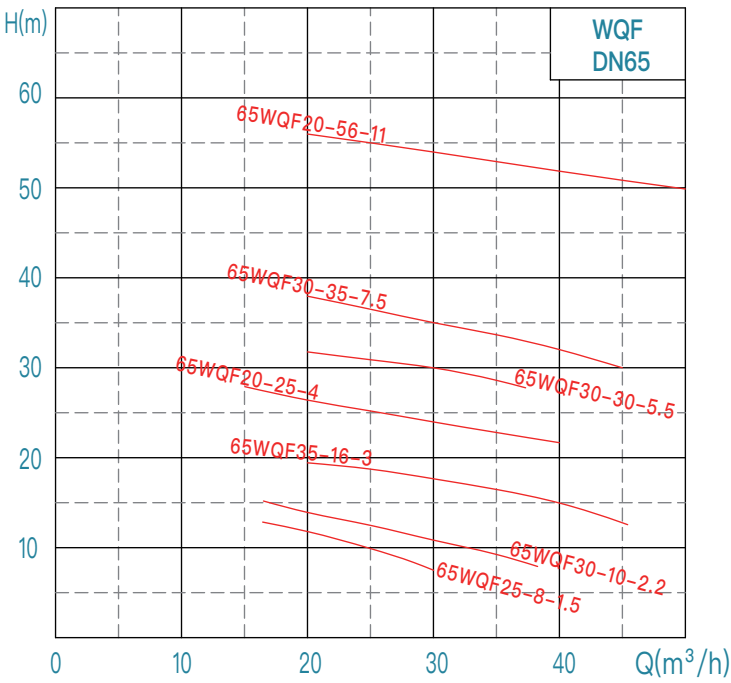
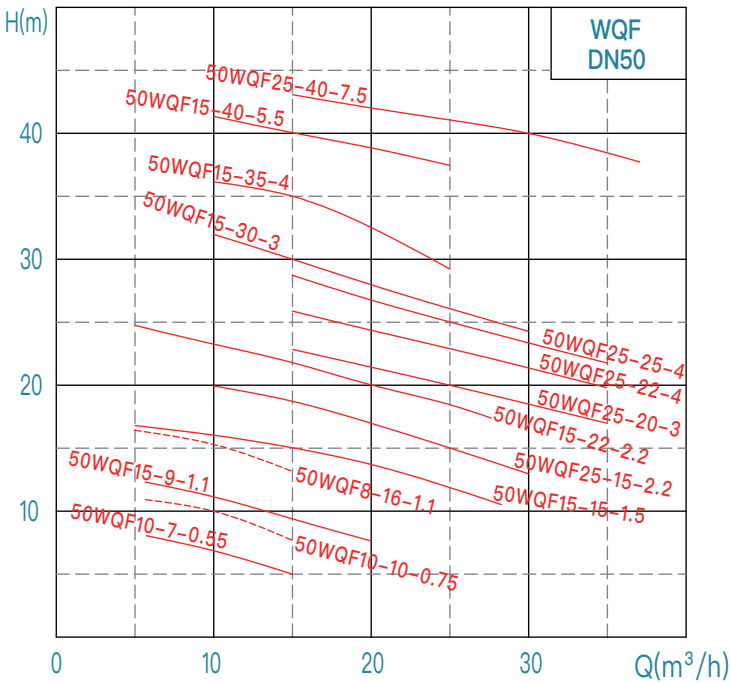
Soft pipe (elbow) mobile installtion

No.	WQF model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Stirring (JY) H、H1、H2 needs to be increased	Weight
		mm	kW	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7		kg
1	50WQF10-7-0.55	50	0.55	2950	220/380	15	135	190	490	420	100	90	95	105	40	21
2	50WQF10-10-0.75		0.75				135	190	490	420	100	90	95	105		21
3	50WQF8-16-1.1		1.1				135	190	490	420	100	90	95	105		23
4	50WQF15-9-1.1		1.1				135	190	490	420	100	90	95	105		23
5	50WQF15-15-1.5		1.5				150	200	530	460	100	95	100	105		28
6	50WQF15-22-2.2		2.2				150	200	530	460	100	95	100	105		31
7	50WQF25-15-2.2		2.2				150	200	530	460	100	95	100	105		31
8	50WQF15-30-3		3		380	20	150	200	550	480	100	105	110	115		36
9	50WQF25-20-3		3				150	225	560	490	110	95	100	105		36
10	50WQF15-35-4		4			15	170	240	600	520	134	120	125	130		49
11	50WQF25-25-4		4				170	240	600	520	134	120	125	130		49
12	50WQF15-40-5.5		5.5				170	240	600	520	134	120	125	130		54
13	50WQF25-40-7.5		7.5				185	320	680	560	140	125	132	140		85
14	65WQF25-8-1.5	65	1.5		220/380	25	160	200	530	460	95	90	100	115	28	
15	65WQF30-10-2.2		2.2		160		200	530	460	95	90	100	115	32		
16	65WQF35-16-3		3		380		160	225	560	490	110	90	100	115	36	
17	65WQF30-22-4		4				165	240	600	520	120	120	130	136	50	
18	65WQF20-30-4		4				165	240	600	520	120	120	130	136	50	
19	65WQF30-30-5.5		5.5				165	240	600	520	120	120	130	136	55	
20	65WQF30-35-7.5		7.5				200	320	680	560	140	125	132	140	85	
21	80WQF40-10-2.2	80	2.2		220/380	30	165	250	550	480	130	105	115	125	33	
22	80WQF40-12-3		3		380		165	250	570	500	130	105	115	125	38	
23	80WQF60-12-4		4				165	240	640	550	150	120	130	140	51	
24	80WQF40-22-5.5		5.5				165	240	640	550	150	120	130	140	60	
25	80WQF65-15-5.5		5.5				165	240	640	550	150	120	130	140	60	
26	80WQF50-25-7.5		7.5				190	320	710	590	175	130	140	150	85	
27	100WQF60-9-3	100	3			380	30	180	250	580	510	140	105	120	135	40
28	100WQF80-8-4		4		175			240	650	560	155	125	137	150	0	51
29	100WQF65-15-5.5		5.5		175			240	650	560	155	125	137	150		62
30	100WQF80-18-7.5		7.5		190			320	730	610	180	130	150	170		85
31	150WQF100-10-5.5	150	5.5		40		200	240	650	560	150	115	130	150		66
32	150WQF130-8-5.5		5.5				275	320	765	645	205	175	200	225	108	
33	150WQF100-13-7.5		7.5			200	320	760	640	200	160	180	195	89		
34	150WQF135-10-7.5		7.5			275	320	885	765	205	175	200	225	111		
35	65WQF20-56-11	65	11		2950	15	240	320	820	680	165	137	145	152	/	106
36	80WQF50-45-11	80	11				240	320	820	680	165	137	145	152		106
37	80WQF50-55-15		15				240	320	850	700	165	137	145	152		115
38	80WQF60-50-22		22				250	440	970	720	177	175	180	185		184

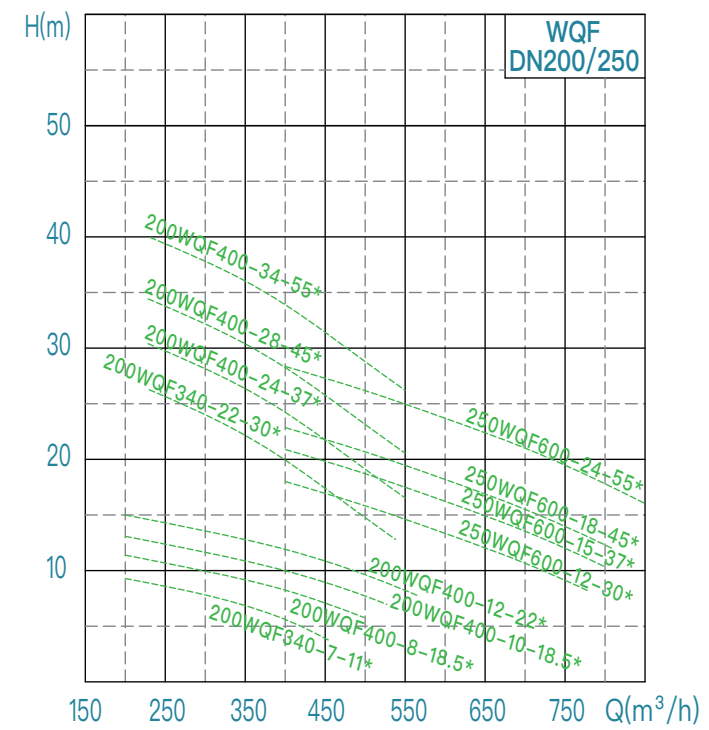
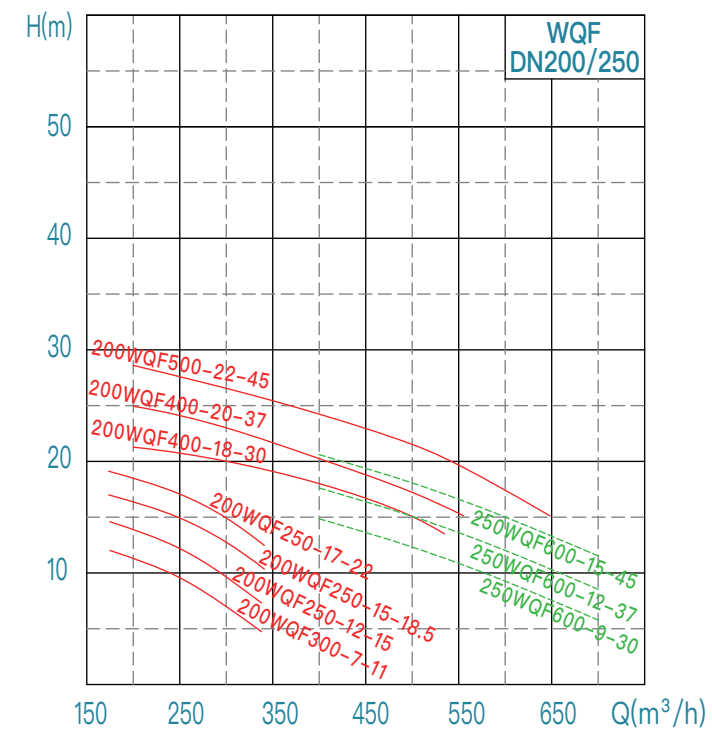
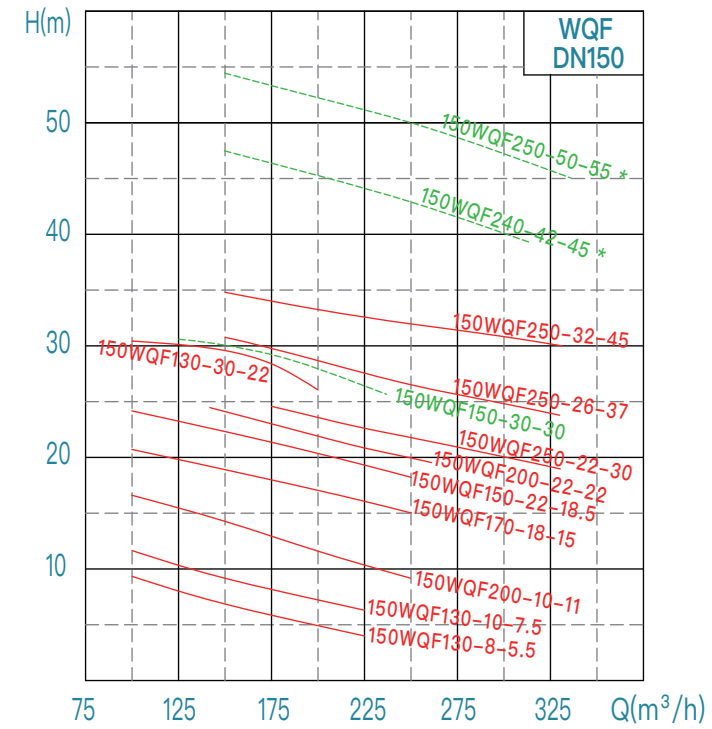
WQF installition dimension table

No.	WQF model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Stirring (JY) H、H1、H2 needs to be increased	Weight			
		mm	kW	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7		kg			
39	100WQF80-25-11	100	11	1475	380	50	280	400	970	780	221	195	212	227	/	195			
40	100WQF150-22-15		15				280	400	970	780	221	195	212	227		212			
41	100WQF80-30-15			280				400	970	780	221	195	212	227		212			
42	100WQF80-35-15			280		400		970	780	221	195	212	227	175					
43	100WQF100-30-18.5			18.5		1475	20	280	400	1120	840	221	195	212		227	247		
44	100WQF100-34-22		22	2950		20		280	400	1170	890	221	195	212		227	264		
45	100WQF80-55-30		30				280	400	1170	890	221	195	212	227		244			
46	150WQF200-10-11	150	11	1475		60	280	500	1100	830	243	223	240	258		195			
47	150WQF170-18-15		15					280	500	1100	830	243	223	240		258	212		
48	150WQF150-22-18.5		18.5						280	500	1150	860	243	223		240	258	255	
49	150WQF200-22-22		22							280	500	1200	910	243		223	240	258	264
50	150WQF130-30-22										280	500	1200	910		243	223	240	258
51	150WQF150-30-30		30				50	350	500	1250		1000	230	242		260	278	383	
52	150WQF250-22-30								350	500	1250	1000	230	242		260	278	383	
53	150WQF250-26-37				37					350	500	1340	1100	230	242	260	278	440	
54	150WQF250-32-45										45	350	500	1340	1100	230	242	260	278
55	150WQF 240-42-45 *		55		500		400	1630	1050	210			260	275	290	572			
56	150WQF 250-50-55 *										500	400				1630	1050	210	260
57	200WQF300-7-11		200		11		1475	70	280	500	1100	830	243	248	265	285	203		
58	200WQF250-12-15				15					280	500	1100	830	243	248	265	285	221	
59	200WQF250-15-18.5				18.5						280	500	1150	860	243	248	265	285	255
60	200WQF250-17-22	22		280	500	1200						910	243	248	265	285	273		
61	200WQF400-18-30	30			320	500			1250	1000	245	250	270	285	391				
62	200WQF400-20-37	37		320		500			1310	1050	245	250	270	285	458				
63	200WQF500-22-45	45				320			500	1370	1150	245	250	270	285	467			
64	250WQF600-9-30	30							400	530	1350	1130	290	218	258	300	400		
65	250WQF600-12-37	37		400	530	1400				1180	290	218	258	300	467				
66	250WQF600-15-45	45			400	530				1400	1180	290	218	258	300	475			
67	200WQF400-8-15 *	200	15	80		420	500	1100		830	285	240	280	310	320				
68	200WQFW400-10-18.5 *		18.5		1150			860	330										
69	200WQF400-12-22 *		22						1200	910					340				
70	200WQF340-22-30 *		30		460	400	1560	980	215	250	280	310	523						
71	200WQF400-24-37 *		37				1640	1060					550						
72	200WQF400-28-45 *	45	583																
73	200WQF400-34-55 *	55	616																
74	250WQF600-12-30 *	250	30		530	400	1570	990	210	265	315	360	550						
75	250WQF600-15-37 *		37				572												
76	250WQF600-18-45 *		45				605												
77	250WQF600-24-55 *		55				638												

WQF performance curve chart (50Hz)



WQF performance curve chart (50Hz)



WQA-QG cutting
submersible sewage pump ►

WQA–QG

Cutting

Submersible sewage pump



Scope of application:

municipal engineering、sanitary sewage、building works、industrial sewage, it can cut and convey ordinary plastic、fabrics、paper、fibers and food residues, etc

Operating Conditions:

Flow Rate: 10 ~ 180 m³/h
Head: 8 ~ 60 m
Medium Temperature: T ≤ 40°C
Medium Density: ≤ 1100 kg/m³; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 10
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

Product Introduction:

The WQA–QG submersible sewage pump with a cutting impeller features a novel, aesthetically pleasing, and highly stable structure. Both the impeller and the cutting disc have sufficient hardness, providing excellent cutting performance for non-tough debris in the medium. Moreover, the cutting disc is equipped with a plum-shaped cutting edge to prevent entanglement at the inlet. (The length of debris that can be passed should not exceed twice the outlet diameter, and the particle diameter requirements are shown in the table.)

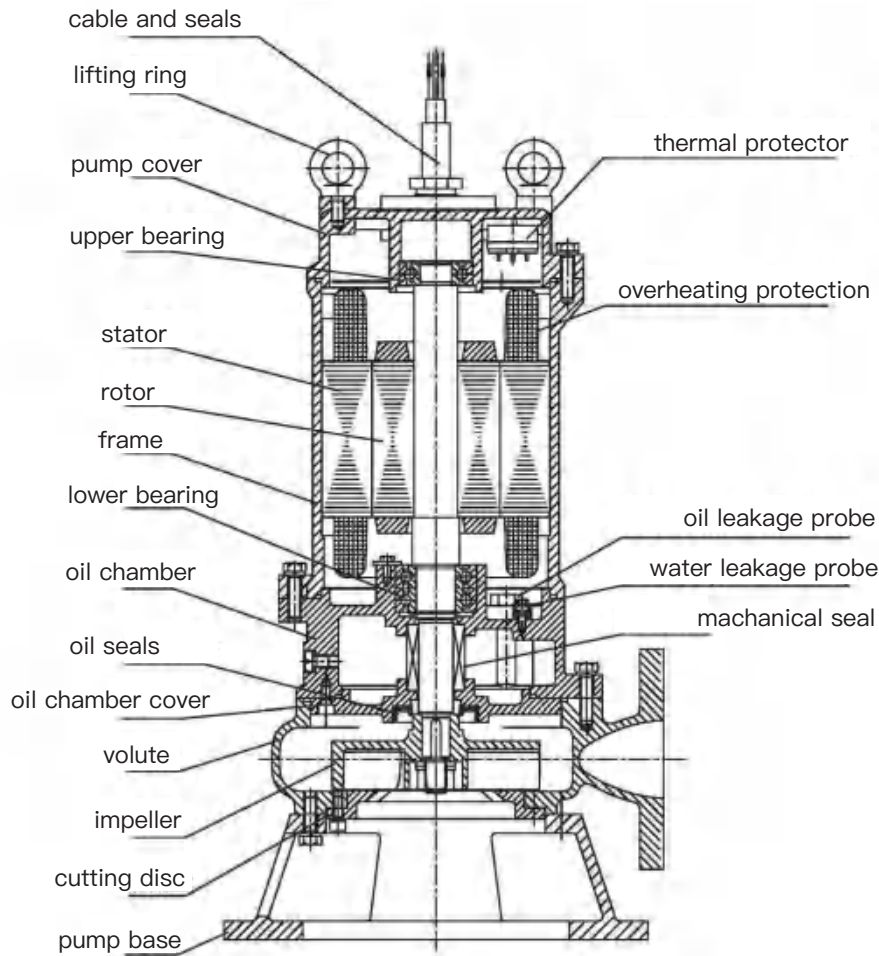
Limitations of Cutting Pumps

Although cutting pumps have strong cutting capabilities, they are not omnipotent. After prolonged use, the cutting head will wear out, leading to a decline in cutting performance. Additionally, the cutting head is prone to being entangled and blocked by long fibers and tough debris. Once this occurs, the cutting pump will fail to operate normally and may even result in damage to the cutting head. This indicates that cutting pumps may encounter difficulties in handling certain types of debris, such as long fibers and tough materials like hair and fabric scraps.



Structure diagram of cutting pump

Structure diagram of WQA-QG products



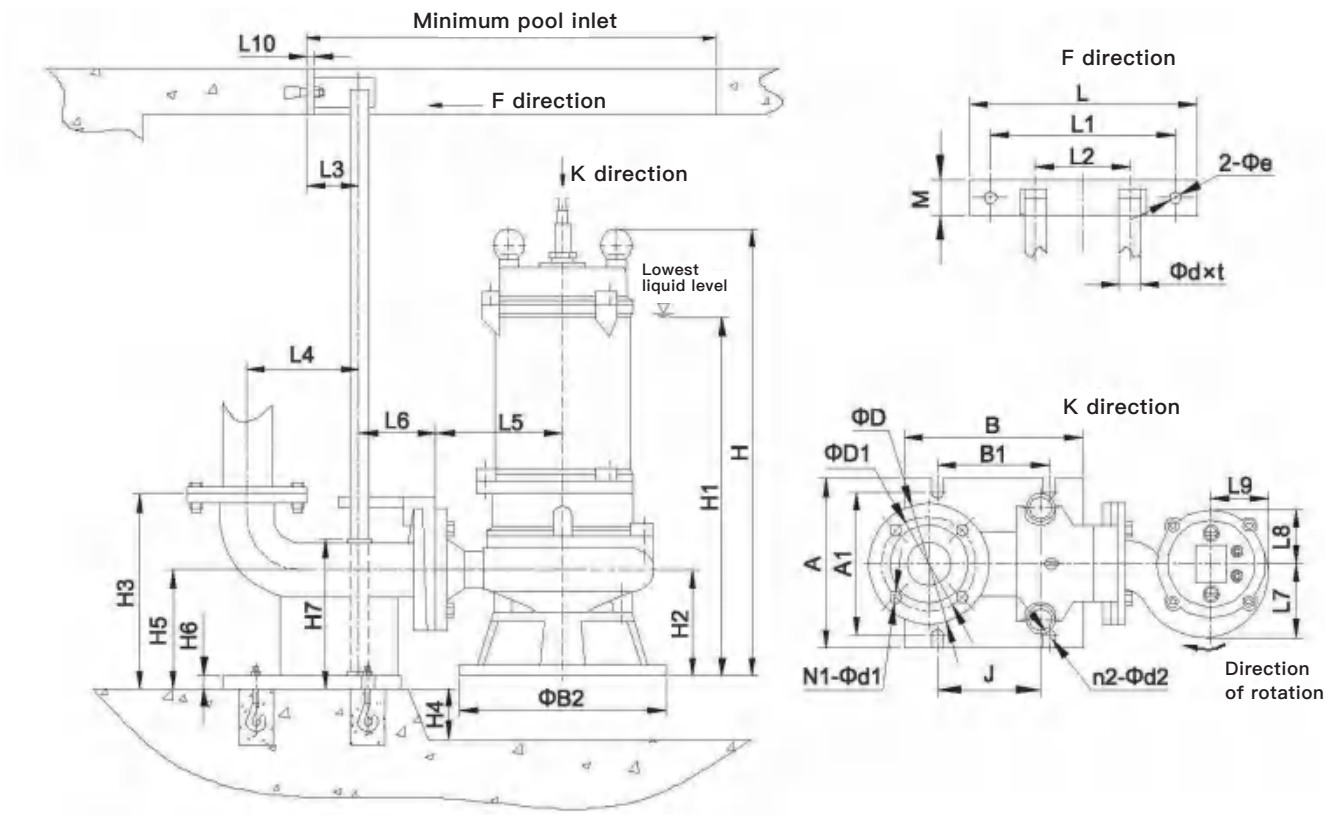
cutting disc /cutting impeller

Main materials and configuration table

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		QT500	/
2	Impeller		QT500	/
3	Cutting disc		QT500	/
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		HT200	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		2Cr13	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/SiC (Q1Q2)	SiC/WC (Q1U1)
11		Spring	Stainless steel	/
12	Rubber parts		(P)	/
13	O-ring		(P)	/
14	External leakage fastener		A2-70	/
15	Protection device (optional)	15kW and below	Order remarks according to customer requirements.	Water leakage probe/Float switch /Overheating protection

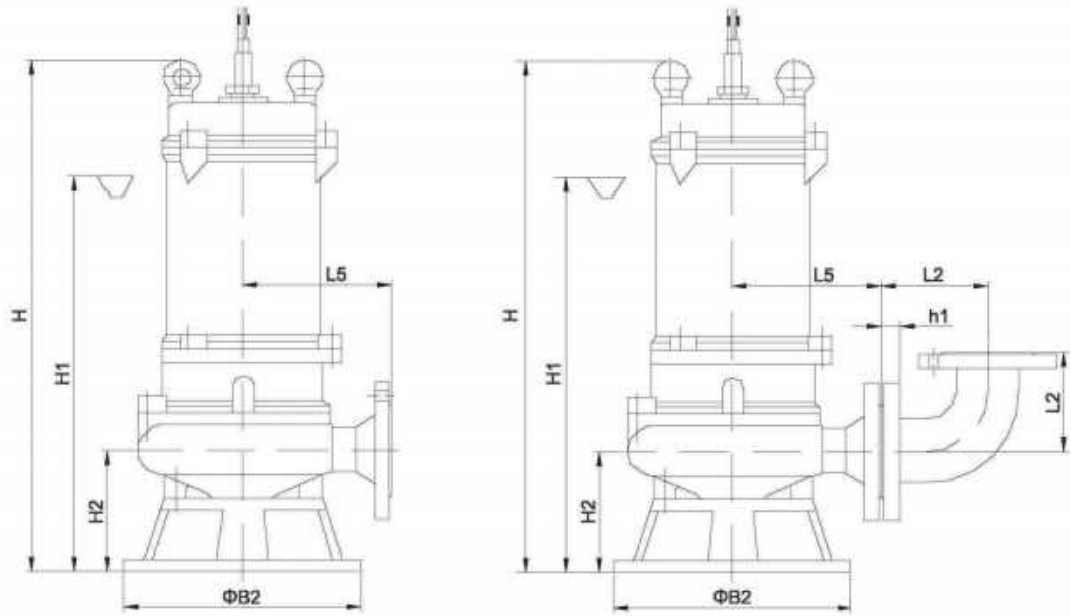
Specification/model of submersible motor cable

Motor power (kW)	Cable model	Standard main cable specification	Star delta	Model of controlling cable
0.55	YZW 300/500V	YZW 4×0.75	/	YZW 4×0.75 (Non-standard)
0.75				
1.1				
1.5				
2.2		YZW 4×1		
3				
4				
5.5				
7.5		YZW 3×1.5+1×1		
11/15-2P				
		YZW 3×2.5+1×1.5		
		YZW 3×6+1×4	2-YZW 3×2.5+1×1.5	YZW 4×1 (Non-standard)



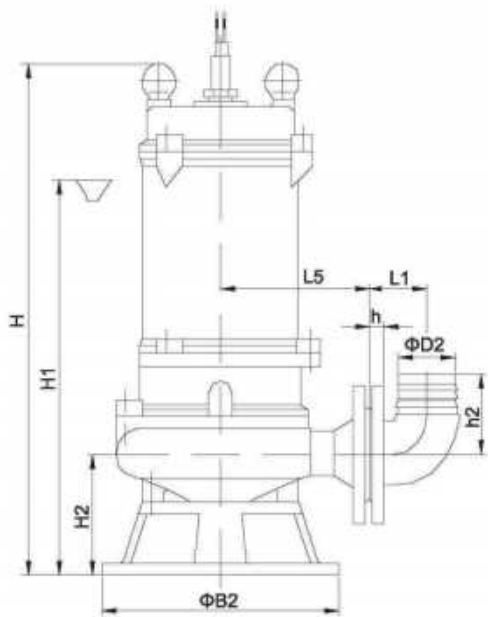
Model DN	Flange Connection Size			Coupling Base Size																				
	D	D1	N1-Φd1	A	A1	B	B1	J	n2-Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φdxt
50	140	110	4-Φ14	197	168	208	130	120	4-Φ19	250	100	165	14	/	265	215	105	42	Φ14	66	135	90	15	Φ33×3
65	160	130		238	190	252	155	135		265		170	16		280	235	125	50		70	155	85	18	
80	190	150	4-Φ19	255	220	225	153	125		310		190	/		315	265	145	53	Φ15	80	155	100	18	Φ48×3
100	210	170		293	265	258	175	140		350		235			18	365	305	170		55	100	200	110	
150	265	225	8-Φ19	400	285	410	300	210	4-Φ12	485	150	300	22	390	410	260	275	60			290	190	20	

Pump appearance and mobile installition



Dimension drawing of
pump appearance

Hard pipe (double flange elbow/joint)
mobile installition

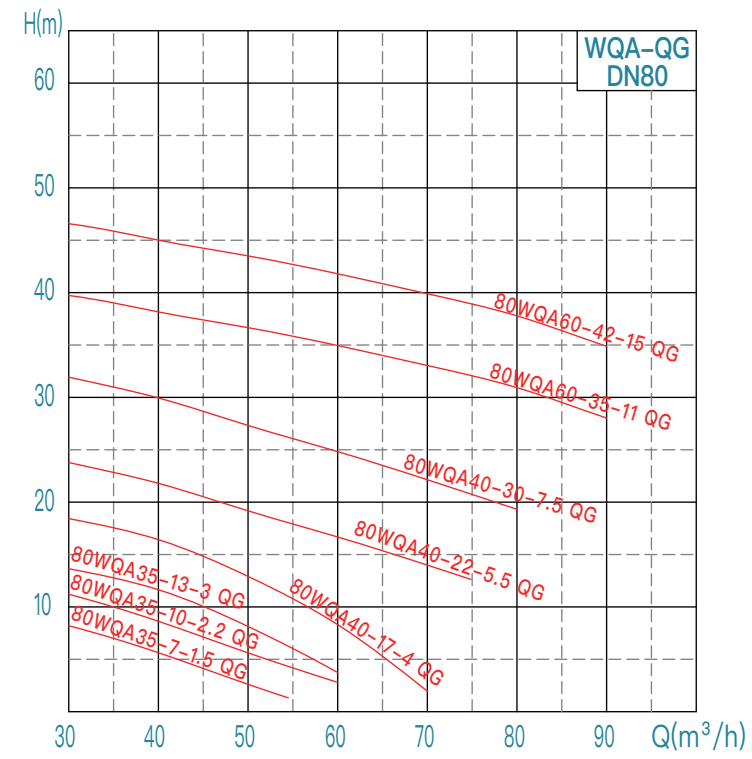
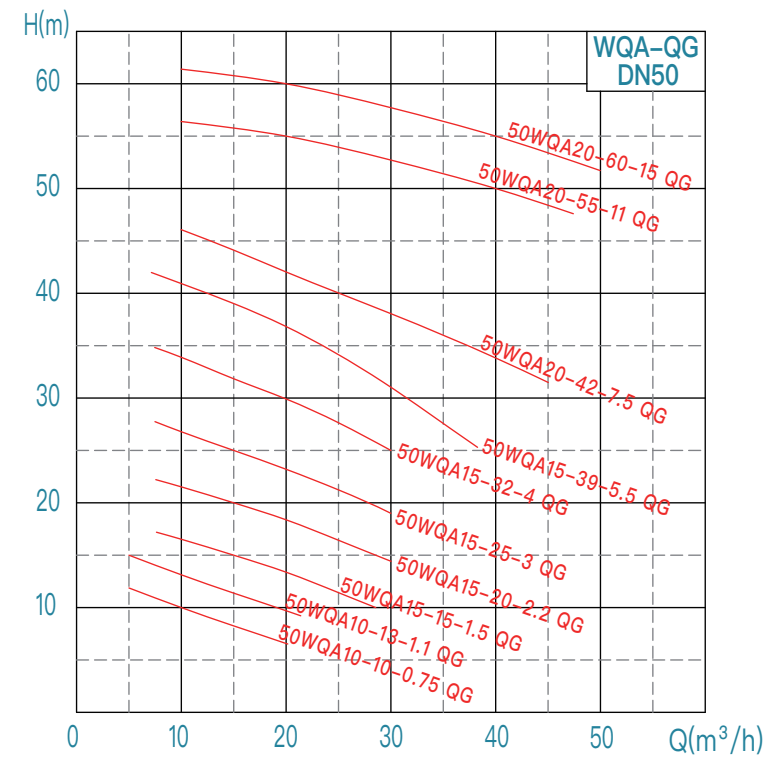


Soft pipe (elbow)
mobile installition

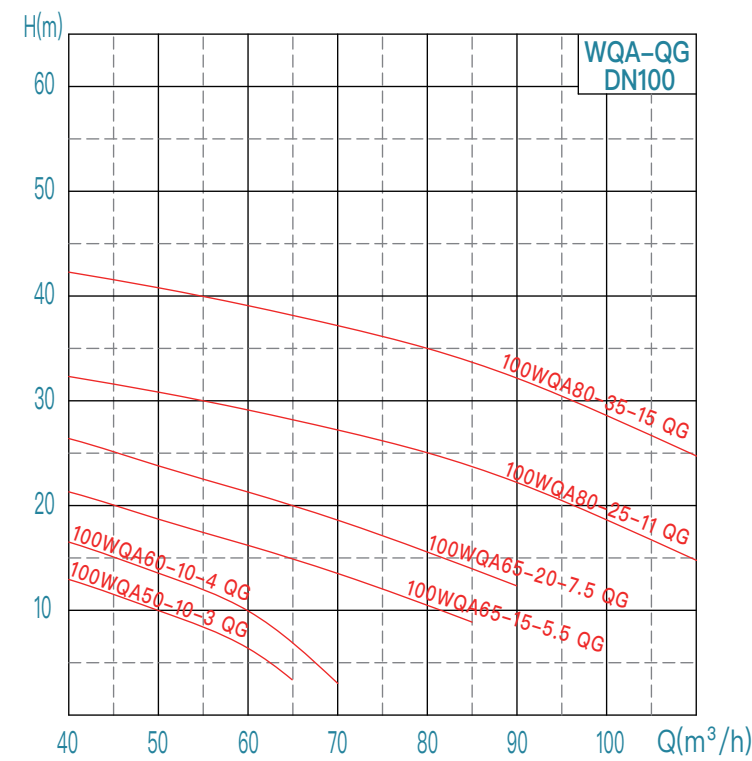
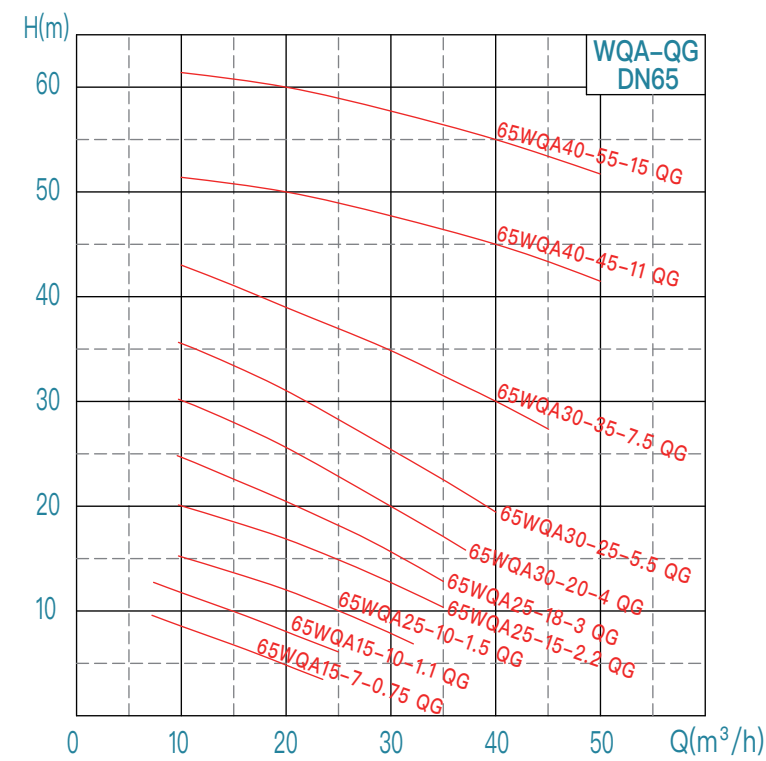
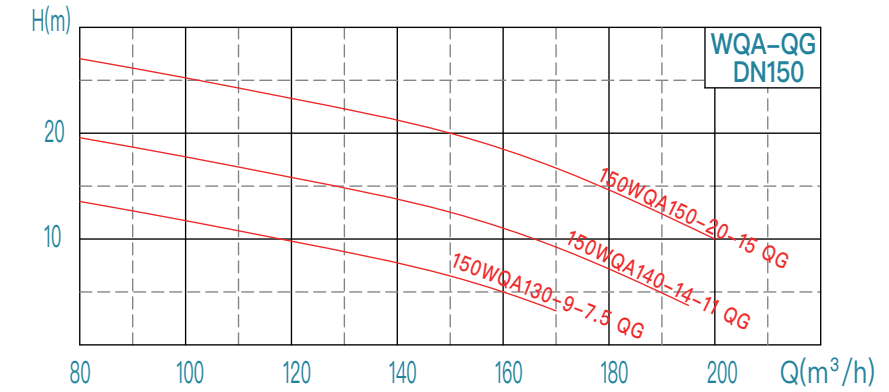
WQA-QG installition dimension table

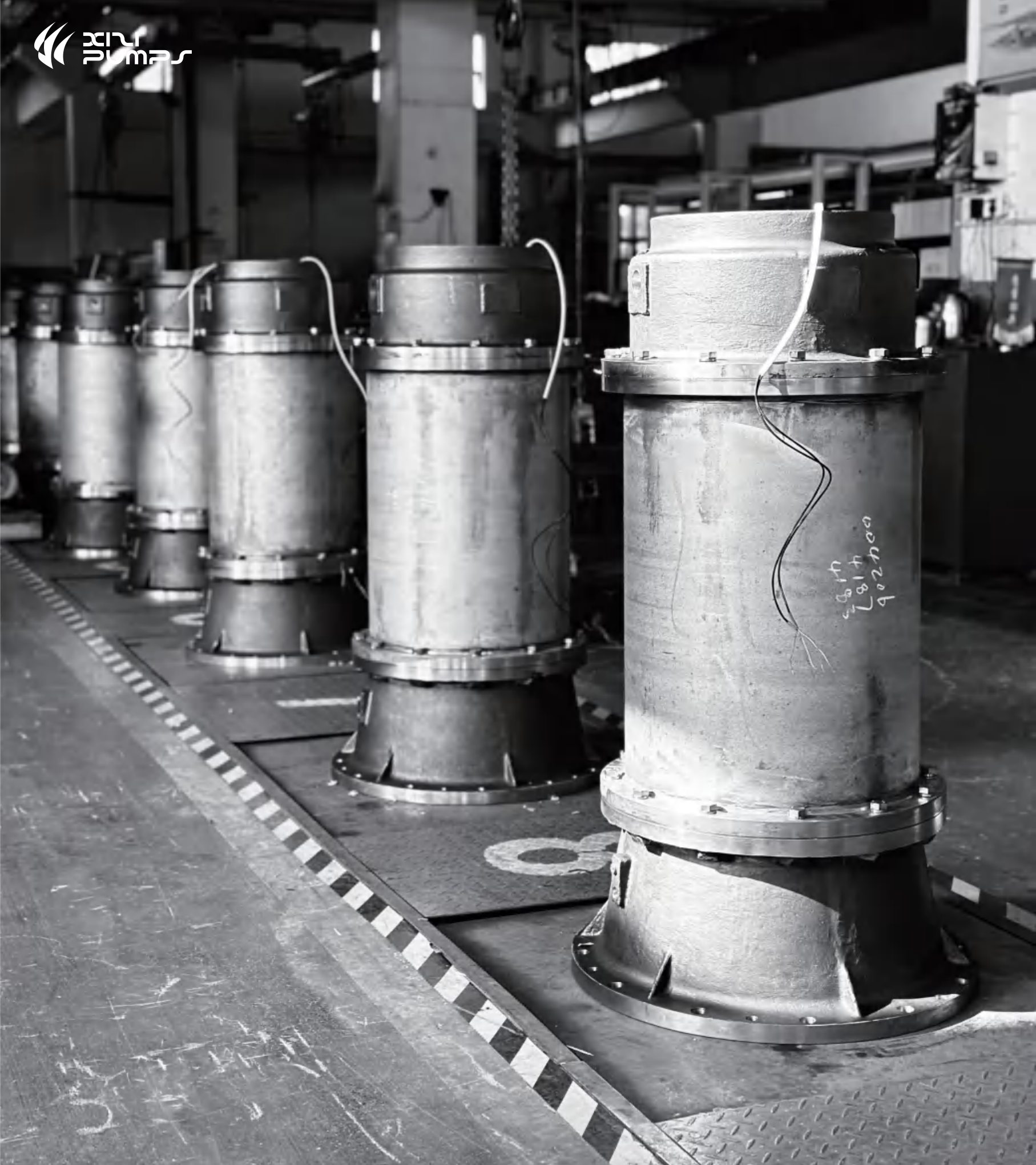
No.	WQA-QG model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight
		mm	kW	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg
1	50WQA10-10-0.75 QG	50	0.75	2950	220/380	15	135	200	460	380	100	82	90	95	26
2	50WQA10-13-1.1 QG		1.1				145	270	485	400	115	100	105	110	28
3	50WQA15-15-1.5 QG		1.5						515	425					32
4	50WQA15-20-2.2 QG		2.2						535	445					37
5	50WQA15-25-3 QG		3						560	460	115	117	120	130	42
6	50WQA15-32-4 QG		4						630	520					59
7	50WQA15-39-5.5 QG		5.5						630	520	135	140	135	145	70
8	50WQA20-42-7.5 QG		7.5				180	320	710	600					90
9	50WQA20-55-11 QG		11						740	630					130
10	50WQA20-60-15 QG		15						740	630					138
11	65WQA15-7-0.75 QG	65	0.75		220/380	15	135	200	460	380	100	82	90	95	26
12	65WQA15-10-1.1 QG		1.1				145	270	485	400	115	100	105	110	28
13	65WQA25-10-1.5 QG		1.5						515	425					32
14	65WQA25-15-2.2 QG		2.2						535	445					37
15	65WQA25-18-3 QG		3						560	460	115	117	120	130	42
16	65WQA30-20-4 QG		4						585	485					59
17	65WQA30-25-5.5 QG		5.5				180	320	630	520	135	140	135	145	67
18	65WQA30-35-7.5 QG		7.5						710	600					90
19	65WQA40-45-11 QG		11						740	630					130
20	65WQA40-55-15 QG		15						740	630					138
21	80WQA35-7-1.5 QG	80	1.5		220/380	40	170	270	495	405	125	110	115	120	40
22	80WQA35-10-2.2 QG		2.2				180	270	525	435					44
23	80WQA35-13-3 QG		3						555	455	130	115	125	135	49
24	80WQA40-17-4 QG		4						585	485					62
25	80WQA40-22-5.5 QG		5.5						610	510					73
26	80WQA40-30-7.5 QG		7.5			25	200	320	645	530	145	140	140	150	95
27	80WQA60-35-11 QG		11						720	610					137
28	80WQA60-42-15 QG		15						750	640	145	140	140	150	144
29	100WQA50-10-3 QG	100	3		380	40	170	270	555	455					49
30	100WQA60-10-4 QG		4				180	270	585	485	130	115	125	135	62
31	100WQA65-15-5.5 QG		5.5						610	510					73
32	100WQA65-20-7.5 QG		7.5			25	200	320	645	530	145	140	140	150	95
33	100WQA80-25-11 QG		11						720	610					137
34	100WQA80-35-15 QG		15						750	640					144
35	150WQA130-9-7.5 QG	150	7.5			50	240	320	670	560	165	145	160	175	128
36	150WQA140-14-11 QG		11						755	645					142
37	150WQA150-20-15 QG		15						785	675					150

WQA-QG performance curve chart (50Hz)



WQA-QG performance curve chart (50Hz)





Ordering, Installation and
After-Sales Service Guidelines ►

Order description

To make the pump you choose more in line with working site requirements, users are warmly welcome to call our technical department.

Please specify when ordering:

- 1. Pump model, flow, head, speed, power, motor power, medium temperature, medium composition, medium density
- 2. Parts material: pump body, impeller
- 3. Installation method, outlet diameter
- 4. Please specify starting method in the order if there are some requirements.
- 5. The standard water pump cable is 9 meters, specify it if other lengths are required.
- 6. The complete set of supply parts is provided according to the determined installation method.
- 7. Necessary parts, optional parts and spare parts need to be ordered separately.
- 8. Follow the principle of "spread before turning" when connecting the spinal canal and the elbow joint on the discharge pipeline.
- 9. Fill in the selection form.

No.	Pump model:	Configuration: standard Optional (specify in the order)
1	Flow: Q= m³/h	Starting method:
2	Head: H= m	Wiring method:
3	Pump immersion depth:h= m	Medium temperature :Normal °C; Max °C; Min °C
4	Outlet diameter: mm	Corrosion component:
5	Insulation class: F(standard) H(optional)	Oxide concentration(PPM):
6	Motor rated power: kW	Sediment content:
7	Voltage on site: V	Maximum particle diameter: mm
8	Pump speed: r/min	Fiber type and length:
9	Frequency: Hz	Ambient temperature:Normal °C; Mak °C; Min °C

Supply list

Supply list		Installation method				
		Automatic coupling installation(Z)	Fixed base installation (P)	Hard pipe mobile installation(Y)	Soft pipe mobile installation(R)	Single pump
Complete supply parts	Pump (9-meter cable)	✓	✓	✓	✓	✓
	Coupling base	✓				
	Gusset plate	✓				
	Coupling rod	✓				
	Pump base (1pc/set)	✓	✓	✓	✓	✓
Essential parts	Guide rod	✓				
	Anchor bolt	✓	✓			
	Expansion bolt	✓				
	Hard pipe (1 pc/set)		✓	✓		
	Soft pipe (1 pc/set)				✓	
Optional parts	Submersible pump monitoring device	✓	✓	✓	✓	✓
	Stainless steel wire rope and rope clamp for lifting pump	✓	✓	✓	✓	✓
	Spinal tube	✓	✓	✓	✓	✓
	Mating flange	✓	✓	✓		✓
	Flange joint		✓	✓		✓
	Gate valve/butterfly valve	✓	✓	✓	✓	✓
	Check valve	✓	✓	✓	✓	✓
	Electric control cabinet	✓	✓	✓	✓	✓
	Terminal box	✓	✓	✓	✓	✓
	Float switch (liquid level)	✓	✓	✓	✓	✓
Spare parts	Impeller	✓	✓	✓	✓	✓
	Bearing	✓	✓	✓	✓	✓
	Mechanical seal	✓	✓	✓	✓	✓
	Seal ring	✓	✓	✓	✓	✓
	O-ring seal	✓	✓	✓	✓	✓

Installation instructions

● installation

1. Check the pump, motor, and fasteners before installation.
2. The weight of the pipeline should not be added to the pump to avoid deformation, vibration, etc.
3. The anchor bolts must be tightened during installation, and the guide rod and coupling gusset must be fixed.

● Start up

1. The liquid level shall not be lower than the minimum liquid level when starting, the minimum liquid level is shown in the installation diagram ▽.
2. Start the motor, check rotation direction. The pump should rotate clockwise looking down from the motor side.
3. (1) For centrifugal type impeller, close the valve and pressure gauge on the outlet pipeline.
- (2) For mixed-flow and axial-flow impellers, open the valve according to the pump model.
4. Start the motor, turn on the pressure gauge. As the speed rises, gradually adjust the valve opening extent on the outlet pipe to achieve the required flow.

● Stop

1. First close the outlet valve, close the pressure gauge, and then cut off the motor power supply.
2. In case of long-term shutdown, it is recommended to lift the pump up, clean it up, put it in a dry place, and keep it properly.

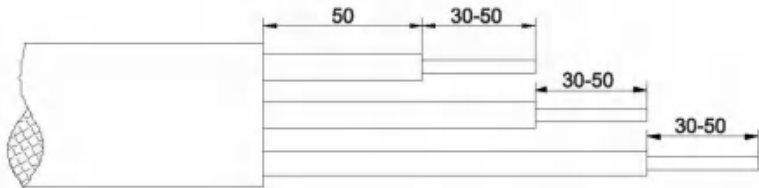
● Maintainance

1. Check the motor current value regularly
 - ① When the ambient temperature is motor rated temperature, the current must not exceed the rated current.
 - ② When the ambient temperature is lower than motor rated temperature, the current value can be slightly increased, please refer to the motor data for details.
 - ③ When the current is abnormal, please find out the cause immediately.
2. After long-term operation, the pump should be shut down for inspection due to the wear of mechanical seals, bearings, impellers, seal rings and other parts. And wearing parts like mechanical seals, bearings, O-rings and seal rings should be replaced. The general inspection cycle is normally one year. The service life of the mechanical seal is determined by the on-site working conditions, and the designed service life is 8000 hours.
3. Check the oil chamber regularly. If oil emulsification occurs, replace 32# engine oil in time. Executive standard: GB443-89

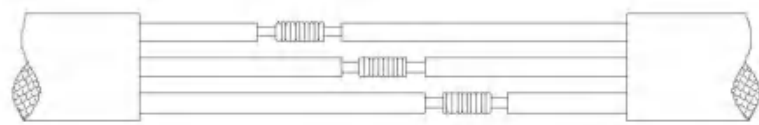
Diving cable connection

When the length of the cable cannot meet users' installation requirements, an external cable is required. The wiring should be operated by a full-time electrician. The cross-sectional area of the cable is determined by the installation length, motor power and starting method. The motor cable and the external cable joints are required to be reliably sealed, insulated, and have certain strength. The wiring process is briefly as follows.

1. Strip the cable on the sewage pump as shown in the figure below, and strip the copper core by 30~35mm long. Wipe it with gauze and copper wire to make it shine. Also strip the copper wire of the external three-core cable by 30~35mm, and wipe it clean with gauze.

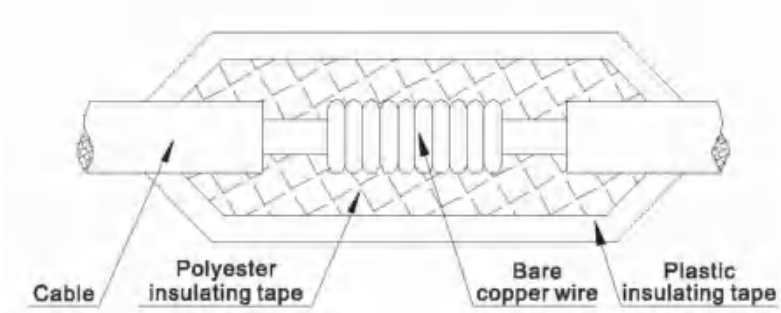


2. As shown in the figure below, insert the stripped copper wire parts of the three-core cable and external cable together, and then use a thin copper wire to bind tightly as shown in the figure, cut off the remaining part, flatten it with scissors, and do not tie your hands. Tie the three wires in the same way.

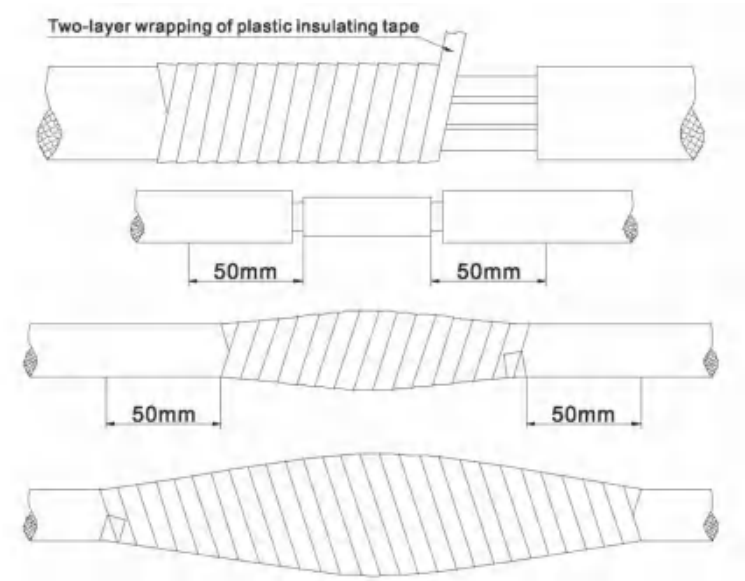


3. Prepare a small iron box or small iron pot that can stick the three wire ends, put solder into the pot and heat it on fire until it melts, and then apply proper amount of soldering oil to the three wire ends, quickly put into the pot and weld it firmly. The surface of the welding head is required to be smooth, free of burrs and false welding. If the welding is not strong or not smooth, it should be re-welded until it meets the requirements.

4. As shown in the figure below, when wrapping with polyester insulating tape, be sure to press half of the first circle (shift wrapping method) for 8-12 layers. After wrapping, wrap two more plastic tape layers for strength protection.



5. As shown in the figure below, wrap the three wires together with plastic tape by two layers. The first layer exceeds the end of each wire insulation layer by more than 50mm, and the second layer exceeds the first layer by more than 50mm. It needs to be tightly wound to exclude air space to the largest extent.



6. After wrapping, take a basin of cold water and completely soak the banded thread in the water. Measure the insulation with a 500V megohmmeter after 12 hours. The insulation should not be less than 50 megohms, otherwise it should be re-wrapped until it meets the requirements.

7. The grounding wire of the submersible electric pump also needs to be wrapped according to the wiring process requirements of the submersible cable.

Note: Customer needs to calculate voltage drop when connecting additional cable. Pay attention to voltage change during operation which should be generally controlled within ±5% of the rated voltage.

Common malfunctions and solutions

Malfunctions	Cause analysis	Solutions
None or Insufficient pump water flow	- Motor reverse rotation - The impeller passage or pipeline is blocked - Tubing resistance is too large - The impeller is severely worn - Gas mixed in the medium - Check valve direction is reversed - Head on customer's site is not enough - Coupling leaks	- Adjust two phase power sequence - Remove impurities, it is best to add a thin grille at the water inlet ① Reduce pipeline bends, increase diameter for long pipe ② Check whether the valve is open ③ Check whether the impeller is scraped - Replace the impeller - Exhaust gas, increase liquid depth to reach the minimum water level requirement "Δ" - Adjust the direction of the check valve - Check whether customer's test method is correct after excluding the above reasons - Check whether the installation is proper and ensure that the guide rod is vertical
The pump fails to start	- Lack of phase - Impeller sticks - Short circuit of winding joint or cable - Stator winding burnt out - Control cabinet error - Low power supply voltage	- Check the circuit, eliminate problem of lack-phase - Clear impurities - Recover after checking with ohmmeter - Replace the winding or stator, it is better to add a temperature measuring element - Check the control cabinet and remove fault - Solve power supply voltage problems
Large fluctuations in discharge pressure	- The liquid level of suction pool is too low - The suction elbow is not tightly sealed, making air entering the pump - Medium temperature is high	- Control the minimum liquid level of the suction pool - Check the pipe connection and eliminate the cause - Reduce medium temperature. If the temperature cannot be lowered, increase the pump minimum liquid level and replace high-temperature motor
Overloading current	- The head of the selected pump exceeds the actual head. Or the head of the device is greatly reduced, indicating that the pump is running at a large flow rate - The density or viscosity of the medium is too high - Bearing is damaged - There are impurities at the neck ring - Power supply voltage is too low	- Reduce outlet pipe valve opening or the outer diameter of small impeller, or replace a pump that matches the actual working conditions on site - Dilute medium - Replace bearing - Clean up impurities - Solve power supply voltage problems
Heating bearing	- Bearing wears - Bearing grease is too little or too much - Grease is degenerative	- Replace the bearing - The amount of grease added is about 1/3~1/2 of the oil chamber cavity - Change grease

Keep improving
Infinite progress !
精益求精，无限进步